
Users Guide

Hammer XMS



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Introduction

This document describes, and provides procedures for using the Empirix Hammer XMS system, designed to provide configurable, carrier-scale monitoring of VoIP signaling and media packet streams. Hammer XMS system capabilities include

- Signaling Analysis
- Signaling and Media Correlation
- Passive VQ monitoring
- Call Analysis
- Centralized Reporting of critical call data and VoIP network metrics.

This document is intended for those users of the Hammer XMS, tasked with monitoring and generating reports of VoIP network performance.

Finding Additional System Information

The following documents and online information are included with the Hammer XMS.

Title	Description
Hammer XMS Release Notes	Most recent changes, resolved issues, and work arounds.
Hammer XMS Hardware Installation Guide	System hardware components, and installation procedures and requirements.
Hammer XMS Software Installation and System Configuration Guide	Instructions for installing/updating system software, and procedures, options, and examples for configuring the system using the configuration script.
Hammer XMS SNMP Guide	Accessing statistics and configuring SNMP alarms and notifications for HXMS .

Title	Description
Hammer XMS CD-ROM Booklet	Software Installation Procedures and requirements.
Hammer XMS Administrators Guide	Creating User accounts, configuring SNMP, the system database, system interfaces, and system operating parameters.
Online Help	Describes all system functions, and procedures for using them. Also includes links to PDF versions of all system guides.

Understanding System Components

The Hammer XMS components include

- One or more probes that monitor media and signaling packet streams.
- A correlation engine that identifies and matches signaling and media call data.
- A system database that collects and stores call data records, supports and manages data searches, data summarizations, system metadata, and control tables.
- A Web user interface that provides access to system data and configuration.

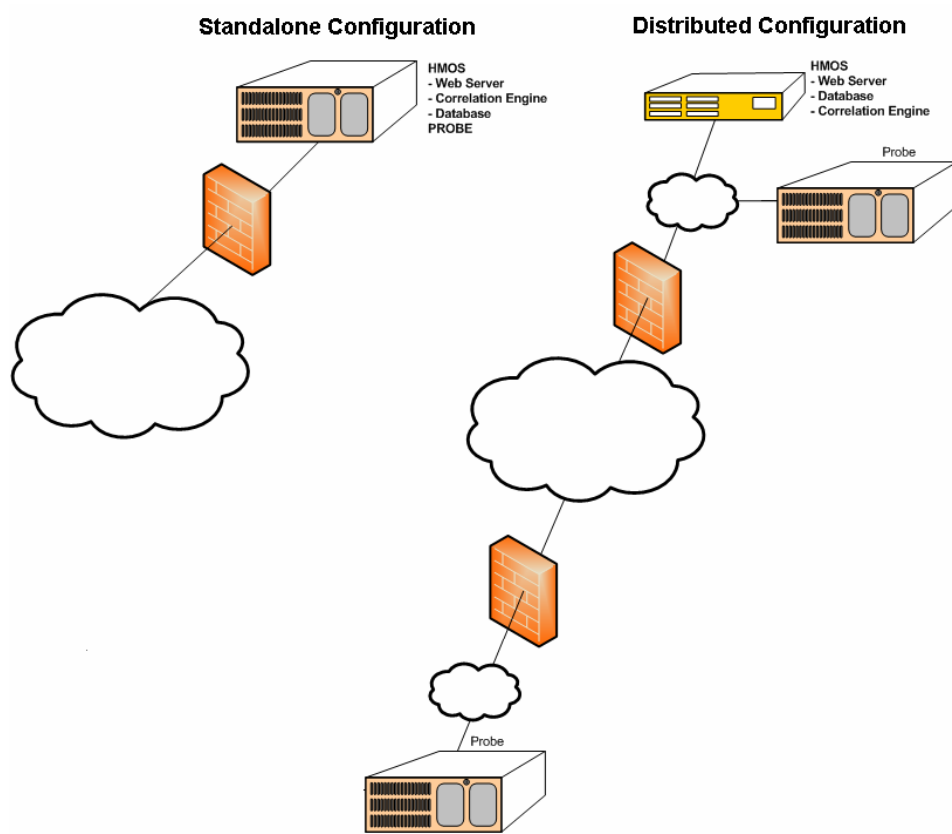
Choosing System Configuration Options

The Hammer XMS is modular, allowing configurations that support a variety of VoIP network monitoring and reporting requirements. The following figure illustrates two of the more common configurations. *Stand Alone*, and *Distributed*.

Stand Alone – In this scenario, the customer needs to capture packet streams from one or more VoIP network points, but has no need to correlate call

data or packet stream metrics from those points. Thus, each probe provides data to a dedicated Hammer Monitor Operations System and Database.

Distributed – In this configuration, multiple probes monitor and analyze packet streams at multiple locations. The resulting data is correlated within one HMOS/database.



Refer to the *Hammer XMS Software Installation and System Configuration Guide* for more information.

Using the Hammer XMS Web Interface

The Hammer XMS Web Interface provides users with centralized access to the Hammer XMS system probe from a remote client, using any of several standard internet browsers.

Insuring Browser Compatibility

Hammer XMS Version 1.5 is known to be compatible with the following Internet Browsers

Browser	Version(s)
MicroSoft Internet Explorer	6.0
Mozilla (Suite)	1.6, 1.7
Mozilla Firefox	1.0
Netscape	7.1

Reviewing User Interface Main Features

The Web Interface provides access to the following main system features, once a user has logged in:

- **Diagnostics** – This feature allows full call data search capabilities, including completed and active calls, filters, tags, and call data update cycles.
- **Reports** – The Reports feature allows the user to generate meaningful call data reports from correlated data.
- **Analysis (if Licensed)** – This feature allows the user to analyze data collected after calls are completed, and subsequently, generate useful reports on that data.

- Administration (requires administrator rights) – The Administration feature allows system administrators to configure probes and interfaces, set node mapping rules, define user accounts, tags, call priorities, assign colors to specific call errors.
- Help – The help function provides access to browser-based web help, and PDF versions of this guide.
- Logout – Logs out the user from HXMS access.

Note: If your system administrator upgrades the Hammer XMS software at any point, and you typically use a mozilla browser (Mozilla, Firefox, or Netscape) your client browser's SSL certificate may not be valid the next time you attempt to connect to the Hammer XMS. If you receive a certificate error message, delete you old certificate as follows:

1. Open your browser's *Tools* menu.
2. Locate the website certificates management utility.
3. Delete your current Hammer XMS SSL certificate.

Using the Hammer XMS

Once installed and initialized, the Hammer XMS is ready to monitor and analyze VoIP packet streams. The resulting data is stored in the Hammer XMS database, from which a variety of reports can be generated.

How To Quick Start Reference

The following table lists the most common Hammer XMS tasks, and where procedural information for each task is located within this guide.

How To	See Page #
Log in to Hammer XMS	13
Creating and using Call Lists	25
Find unusually long active calls	30
Find active test or orphaned ¹ calls	29
Search for calls by time interval	36
Search for calls by IP address	37
Search for calls using filters	38
Search for calls by error code	39
Search for calls by predefined tag	42
View call details	44
Export data to other applications	56
Format and print data	63
Email captured data	63
Use the Analysis feature	64
Create a Report	78

¹ Orphaned calls are calls that show as active though they are no longer connected.

Logging In

The Hammer XMS system administrator is responsible for creating user accounts for system users, including user name and password. If you have not received a user name and password contact your Hammer XMS System Administrator.

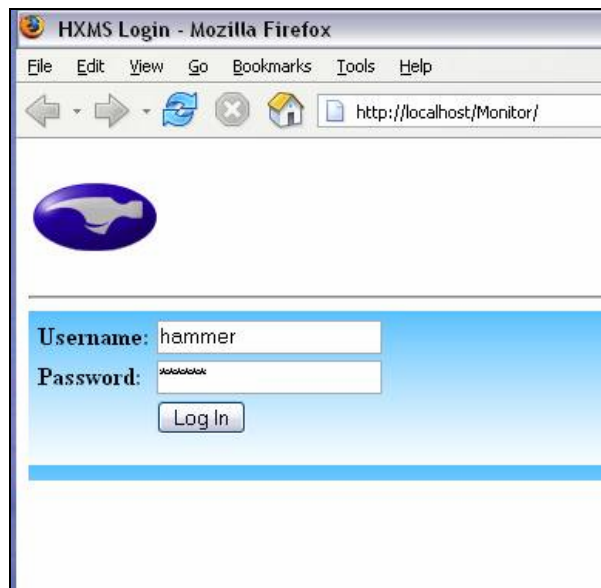
Steps

To log into the Hammer XMS server:

1. Open an Internet browser, and enter the IP address or name of your Hammer XMS system, and press <Enter>.

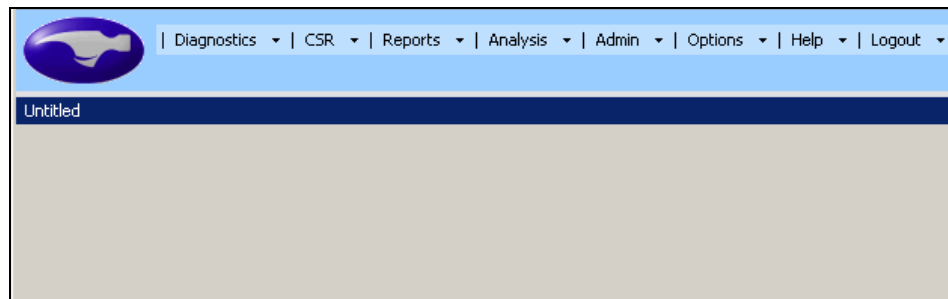
The login dialog, shown in the following figure, will appear.

Note: You will need to know your assigned login and password in order to log in to the Hammer XMS. If you don't know the IP address, login or password, contact your system administrator.



2. Enter your user name and password, and click the *Log In* button.

The Hammer XMS main page will appear as shown in the following figure.



2. At the Hammer XMS main page, click one of the following links:
 - Diagnostics
 - CSR
 - Reports
 - Analysis
 - Admin (system administrator only)
 - Options
 - Help
 - Logout

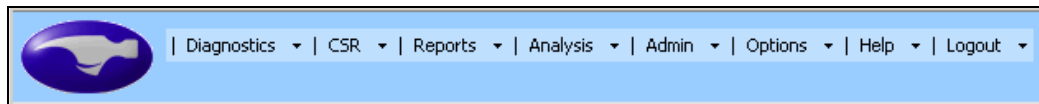
Logging Out Securely

It may be possible for some browsers to cache and share Hammer XMS user credentials across multiple instances of that browser.

This could allow an unauthorized user at your workstation to access the Hammer XMS server through any existing open browser instance.

To prevent this from occurring, always follow these steps when closing the Hammer XMS User Interface:

1. At the top of the Hammer GUI page, click the *Logout* link shown in the following figure.



2. Close ALL instances of your browser, whether or not you have used them to access the Hammer XMS server.

Using HTTPS

Hammer XMS comes configured to provide a Secure Sockets Layer (SSL) connection between the web server and a clients web browser, during login.

The system can also be configured, by the system administrator, to use SSL for all data transactions between client and server.

SSL is a protocol designed to enable secure communications on an insecure network such as the Internet. SSL provides communications integrity through encryption of communications, supplemented by strong authentication using digital certificates.

Typically when accessing a website that uses an SSL connection the only noticeable change is a lock icon in the bottom right of the browser window. That icon means that SSL socket 443 communication is in use and that all communications are secure while that icon is present.

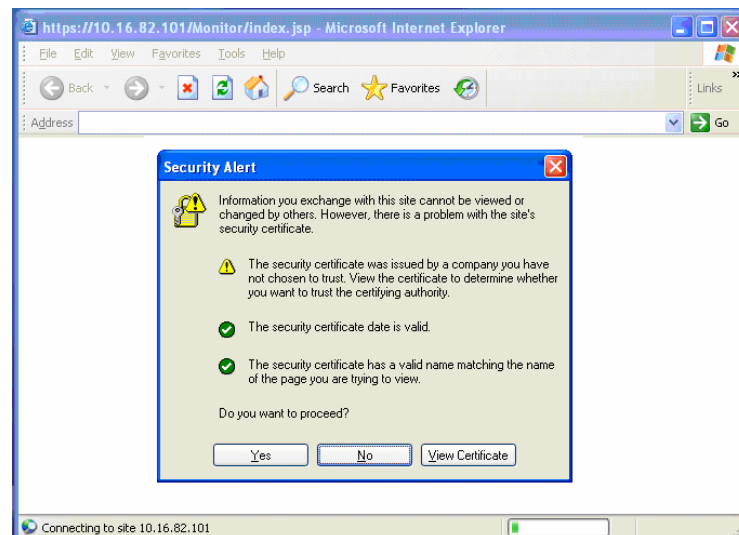
Using this HTTPS, the Hammer XMS creates a self-signed digital certificate that the user accepts as valid. In addition, the user has the option of *always* accepting the certificate.

Installing the Self-signed Certificate

Using HTTPS, each client that connects to the Hammer XMS server has the option of automatically accepting the authentication certificate, or accepting it at the beginning of each client session.

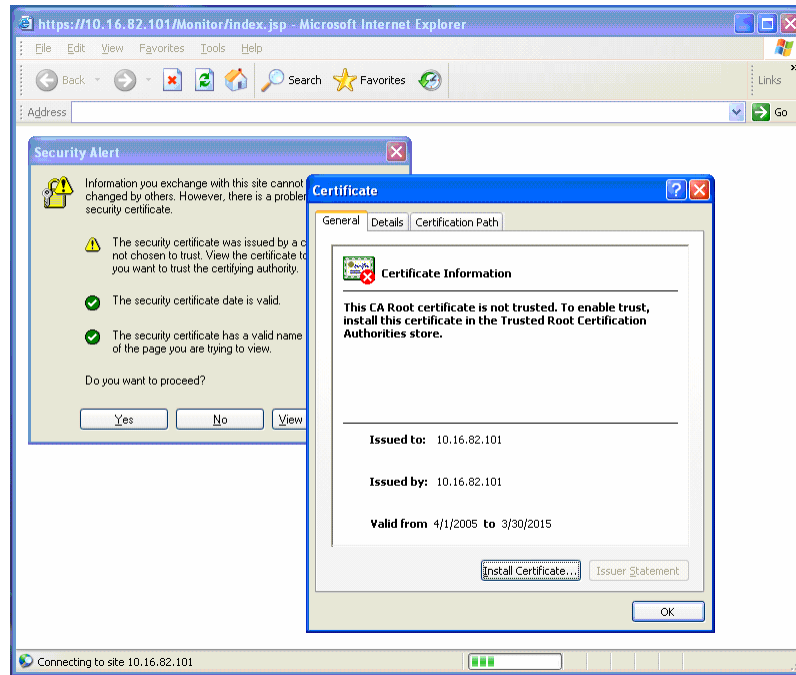
To install the Self-signed Certificate:

1. Open an Internet browser, and enter the IP address for the server. A security alert message will appear, as shown in the following figure.



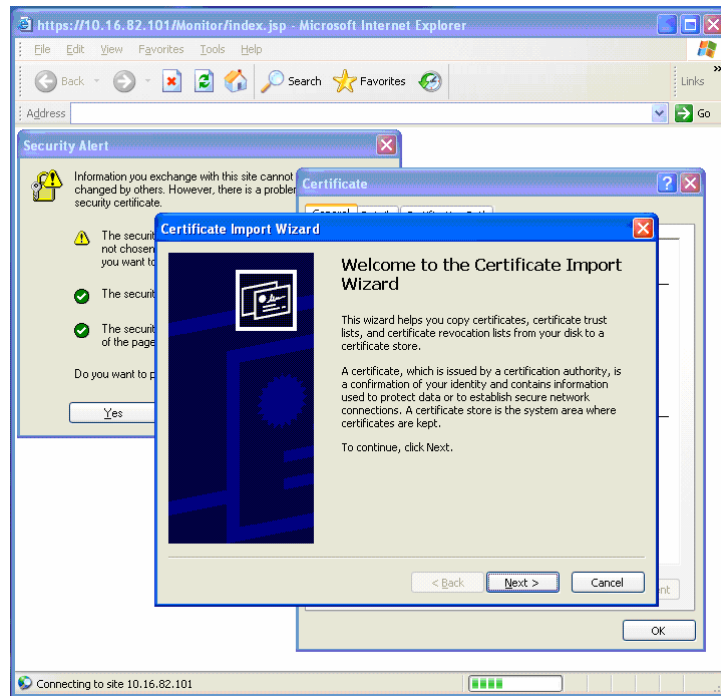
2. Click the *View Certificate* button and click on the *Install Certificate...* button.

The Certificate dialog appears as shown in the following figure.



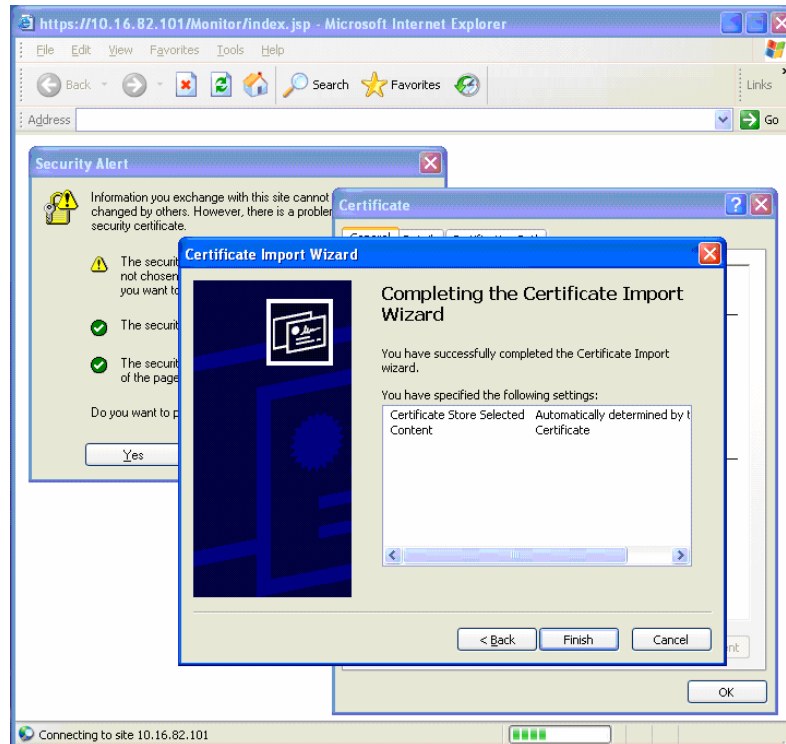
3. Click the *Install Certificate...* button. This stores the digital certificate inside the browser and the user isn't prompted again.

The *Certificate Import* dialog will appear.



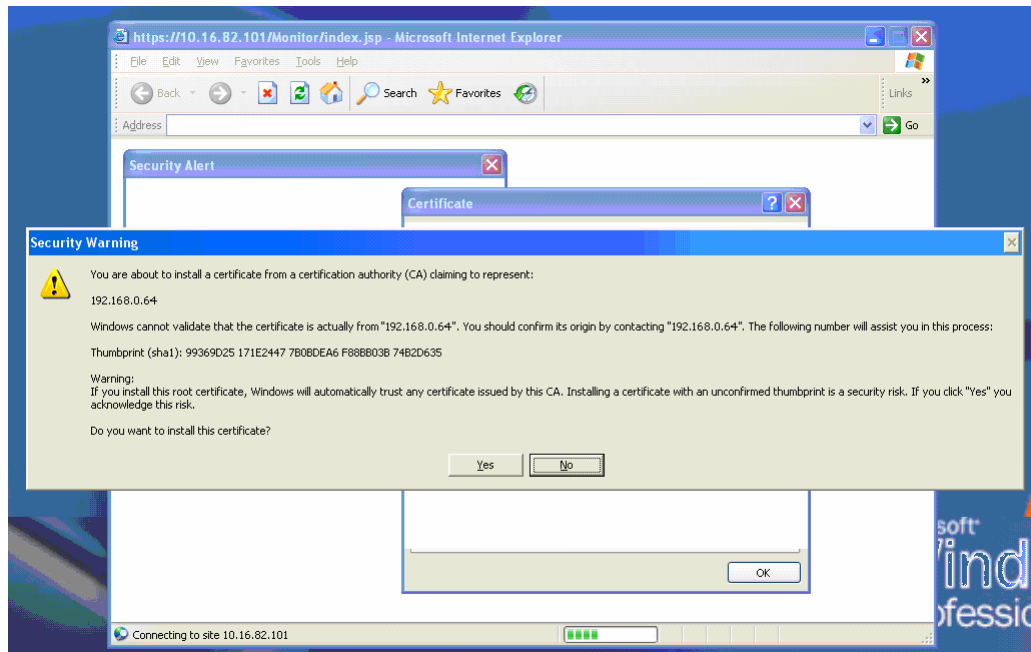
4. Click *Next*. The Certificate Import Wizard dialog appears.
5. Click *Next*.

The Completing Import Wizard dialog appears.



6. Click *Finish*.

The Security Warning dialog will appear, as shown in the following figure.

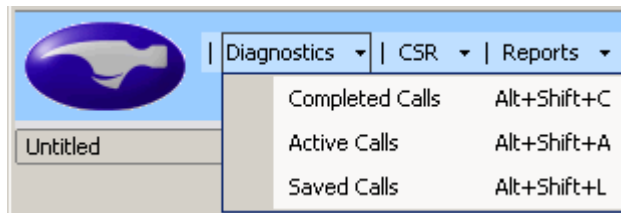


7. Click the Yes button. The HXMS main page will display.

Diagnosing Call Problems

Use the Hammer XMS Diagnostics page to help identify failed calls, connect or disconnect failures, call duration, connect latency, or call originated error codes.

Clicking on the *Diagnostics* link, shown following, displays a drop-down menu with the options *Completed Calls*, *Active Calls*, and *Saved Calls*.



Using Completed and Active Calls

Select *Completed Calls*, shown in the first of two following figures, to search for calls that are stored in the Hammer XMS database. Two types of completed call searches are possible; Basic and Refined. See *Using Basic and Refined Search Modes*, below, for more information.

Select *Active Calls* to search for calls identified as active by the Hammer XMS correlation engine (see second figure).

Important! Keep in mind that *Completed Calls* and *Active Calls* modes use the text search field somewhat differently. In *Completed calls* mode, a user can enter the search text of “1”, pick *OPC* from the *Within* drop-down window, and click search. The data returned for completed calls will return all calls that have an OPC that contains 1. The same is true for DPC.

In *Active Calls* mode, no distinction is made between *OPC* / *DPC* and IP addresses, only between source and destination addresses.

Thus, entering “1” for OPC will return both OPCs AND IP addresses that contains 1 including IP addresses such as 192.168.0.20 and OPC addresses like 1.1.1. The same is true for *DPC* and *Destination IP* options. *Either OPC/DPC* works the same as *Either IP*.

Source / Destination-Type Qualify Search Search Data Search Filter Live Update

Completed Calls - 1

Start Live Update 5 sec (s)

Subscriber Phone Begins With using the filter - No Filter - Search

☒ Last Hour ☐ Today ☐ Last 2 Days ☐ This Week **Refined Search**

Time	Subscriber Name	Source Address	Subscriber Phone	Destination Name	Destination Address	Destination Phone	Duration	Fail Reason
Wed Feb 01 14:46:08 EST 2006		64.74.46.166	41447506506		198.31.145.250	3954011532142670	00:00:00	
Wed Feb 01 14:46:08 EST 2006		172.18.4.3			172.18.114.68		00:00:00	
Wed Feb 01 14:46:08 EST 2006		10.77.22.32			10.77.24.40	9408296259	00:00:00	
Wed Feb 01 14:46:08 EST 2006		64.74.46.166	41447506506		198.31.145.250	3954011532142670	00:00:00	
Wed Feb 01 14:46:08 EST 2006		64.74.46.166	41447506506		198.31.145.250	3954011532142670	00:00:00	
Wed Feb 01 14:46:08 EST 2006		172.18.142.2			172.18.114.48		00:00:00	

Active Calls - 1

Start Live Update 5 sec (s)

Subscriber Phone Contains using the filter Call Duration (s) > -1 Search

Time	Subscriber Name	Source Address	Subscriber Phone	Destination Name	Destination...	Destination Phone	Duration	Priority
Tue Jan 31 16:39:15 EST 2006		172.18.142.2			172.18.114.4i		00:00:00	4
Tue Jan 31 16:39:15 EST 2006		172.18.142.2			172.18.114.4i		00:00:00	4
Tue Jan 31 16:39:15 EST 2006		172.18.4.3			172.18.114.6i		00:00:00	4
Tue Jan 31 16:39:14 EST 2006		70.184.192.36			70.191.98.46		00:00:01	4
Tue Jan 31 16:39:14 EST 2006		aah/1@mta001311240.			aah/1@mta00		00:00:01	4
Tue Jan 31 16:39:14 EST 2006		70.184.192.36			24.248.113.1-		00:00:01	4
Tue Jan 31 16:39:14 EST 2006		ds/ds1-11/17@gainvoig			ds/ds1-11/176		00:00:01	4
Tue Jan 31 16:39:14 EST 2006		70.191.98.65			70.184.192.36	2646750	00:00:01	4
Tue Jan 31 16:39:14 EST 2006		aah/1@mta001311240i			aah/1@mta00		00:00:01	4
Tue Jan 31 16:39:14 EST 2006		70.184.192.36			24.248.113.1-		00:00:01	4
Tue Jan 31 16:39:14 EST 2006		ds/ds1-29/17@gainvoig			ds/ds1-29/176		00:00:01	4
Tue Jan 31 16:39:14 EST 2006		70.184.192.36			70.191.2.130		00:00:01	4
Tue Jan 31 16:39:14 EST 2006		aah/1@mta0013112db.			aah/1@mta00		00:00:01	4
Tue Jan 31 16:39:14 EST 2006		70.184.192.36			24.248.114.1i		00:00:01	4
Tue Jan 31 16:39:14 EST 2006		ds/ds1-11/24@prnscvok			ds/ds1-11/246		00:00:01	4
Tue Jan 31 16:39:14 EST 2006		192.168.0.120	SipPhone02319		192.168.0.12	SipPhone02319	00:00:01	4
Tue Jan 31 16:39:14 EST 2006		192.168.0.120	SipPhone02318		192.168.0.12	SipPhone02318	00:00:01	4
Tue Jan 31 16:39:14 EST 2006		70.191.10.41			70.184.192.36	18002201351	00:00:01	4
Tue Jan 31 16:39:14 EST 2006		aah/1@mta0013112db.			aah/1@mta00		00:00:01	4
Tue Jan 31 16:39:14 EST 2006		192.168.0.120	SipPhone02317		192.168.0.12	SipPhone02317	00:00:01	4
Tue Jan 31 16:38:56 EST 2006		10.77.22.32			10.77.24.36	9408296243	00:00:19	4

Active calls: 14 Show Detail

Use the *Completed Calls* and *Active Calls* pages to:

- Select *Basic* or *Refined* search mode (completed calls only).
- Search for calls by time interval, Subscriber or Destination ID (including IP Address or telephone number), or type of call problem.
- Search by *Tag* definition (License dependent).
- Perform call analysis on individual calls.

- Set *Live Update* parameters.
- Select Filters.

Searches performed using the Diagnostics pages will return a maximum of the 6000 most recent call records, or 6000 of the oldest call records, if the chronological order is reversed. All other search criteria will occur within the 6000 oldest or newest call records. The number of returned call records will display in the bottom left of the search window.

If less than 6000 records exist, the numeric display box will display in green, showing the exact number of records returned, as shown below. If more than 6000, the numeric display box will be yellow.

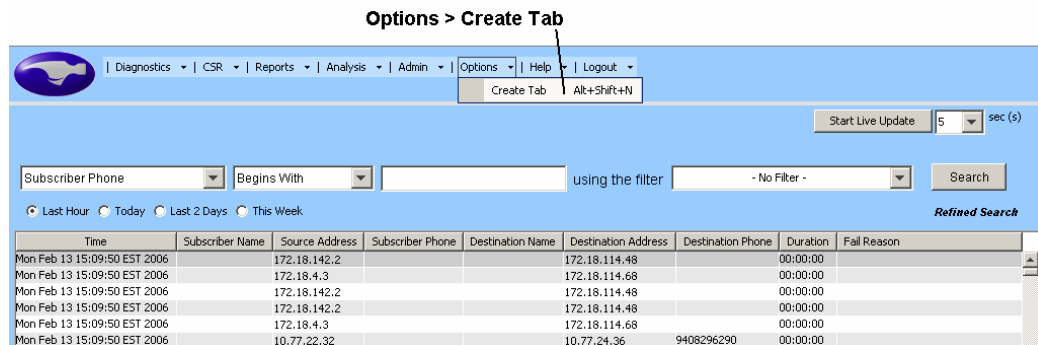


Creating and Using Tabbed Views

Release 1.5 of Hammer XMS provides a tab feature for creating and switching between multiple data pages and views.

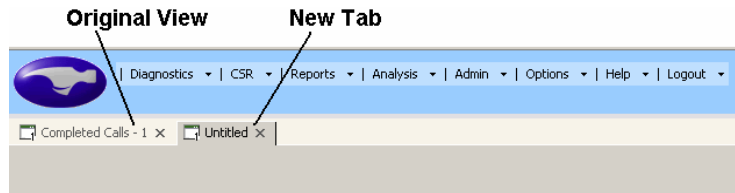
To create and use *tabbed* views:

1. With any view already displayed, click the *Options* link to display the options menu, as shown in the following figure.

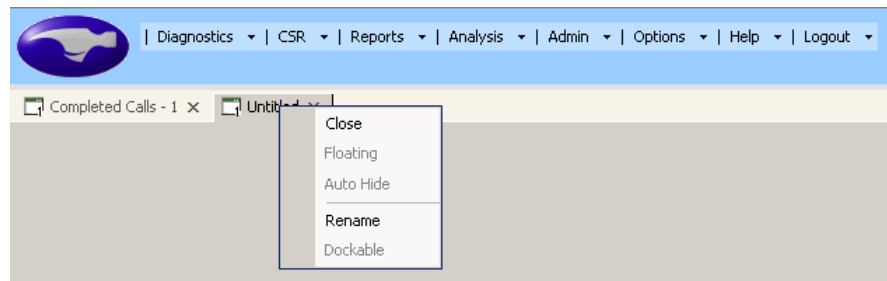


2. Click *Create Tab*.

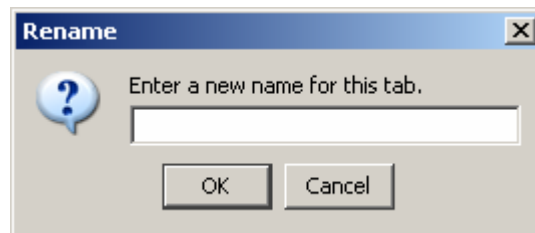
A new blank tabbed page will appear, as shown in the following figure. Gathering additional data with the blank view showing, will result in the data appearing on this new page, allowing you to navigate between multiple views simply by clicking on the appropriate tab.



3. Right-click any tab to open the tab options menu, shown following.



4. Click *Rename* to open the Rename dialog, and type the desired name for the new tabbed page.



5. Click *OK*.

Creating and Using Call Lists

In addition to *Completed* and *Active* call analysis, Hammer XMS Release 1.5 provides a new *Call List* utility that allows you to save individual calls for later analysis.

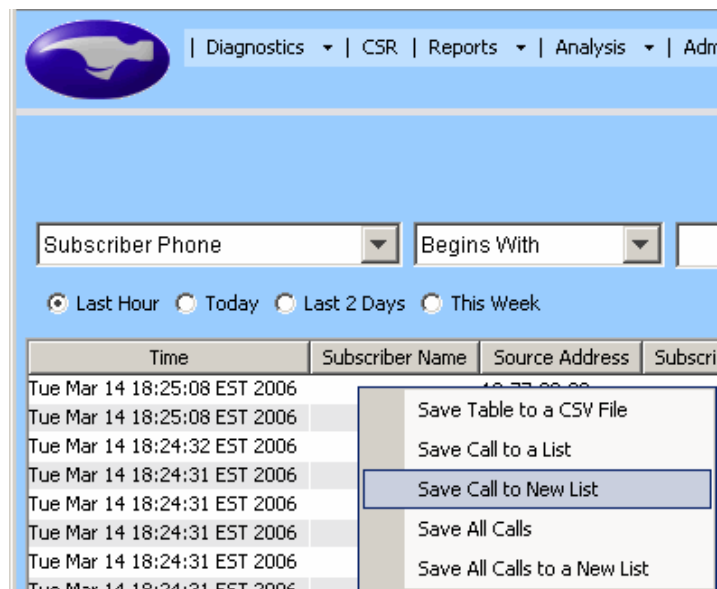
Using this feature, you can:

- Create, delete and rename call lists for calls you wish to save for later analysis.
- Add calls to, and delete calls from each list you create.

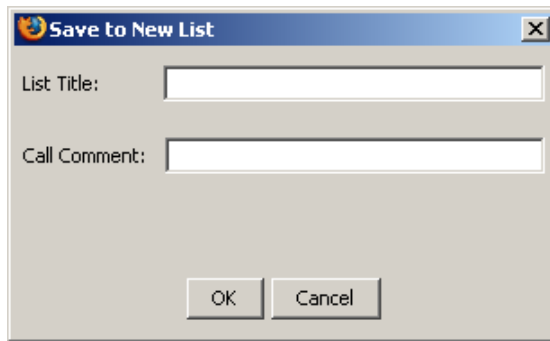
To save a call to a new call list:

1. After performing a call search, and displaying the returned data, right-click on any call you wish to save from a call data search, to display the *call options* menu.

The following dialog will display.



2. Select *Save Call to New List*. The *Save to New List* dialog will display.



3. In the *List Title* text field, enter a name for the new list.
4. In the *Call Comment* text field, add any pertinent comment regarding the call you're saving.
5. Click *OK*.

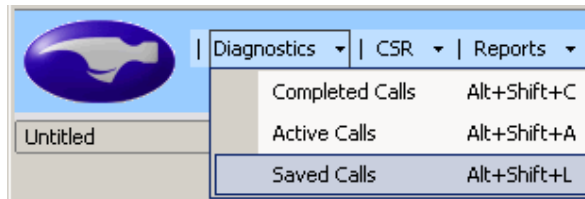
To *Save All Calls to a list*, or *Save All Calls To a New List*:

1. Select *Save All Calls* to add a maximum of 512 calls to an existing list. Or select *Save All Calls To a New List* to add a maximum of 512 calls to new list
2. In the *List Title* text field, enter the list name. In the *Call Comment* text field, add any desired comment regarding the call you're saving.

Note: The *Save All Calls*, and *Save All Calls To a New List* features will only save calls when the maximum number of calls returned by a search is less than 512. If you attempt to use either of these functions for a call list that contains more than 512 calls, you will get the error message "*The number of calls to be saved exceeds the limit allowed. The Limit allowed is 512.*"

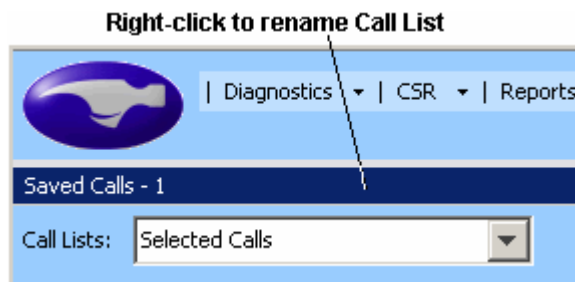
To access previously saved calls:

1. From the menu bar, at the top of the main page, select *Diagnostics*, and select Call Lists, as shown in the following figure.



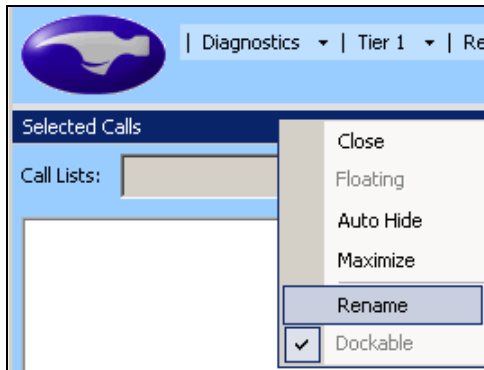
Note: The first time you use the Call List feature, a default call list, *Selected Calls*.

2. Right-click the dark blue line above the Call Lists drop-down menu to rename the call list.

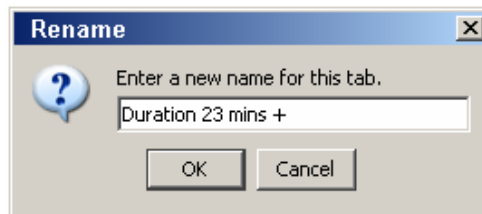


The options menu will display, as shown, providing options:

- Close - Closes the current window.
- Auto Hide - Minimizes the current window.
- Maximize - Expands the current window
- Rename - Allows you to rename the current window.



3. Select *Rename*. The *Rename* dialog displays, as shown in the following figure.
4. In the text field, enter a name for the call list, and click *OK*. For example, if we were collecting calls that exceeded 23 minutes in duration, we might call this list *Duration 23 Mins +*.



Performing Call Data Searches

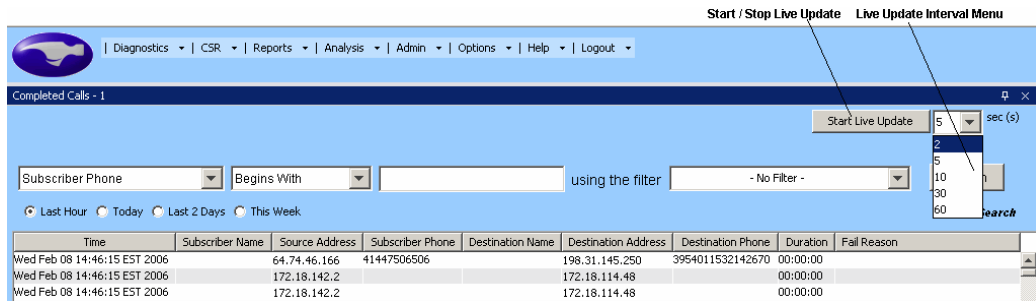
Hammer XMS call data is optimized to reduce search time to a minimum, particularly when *Begins With/ Equal To/ Contains* qualifiers and text are provided. The Diagnostic window provides a variety of search controls for retrieving specific call data. Search features include:

- Live Update
- Completed or active calls
- Time interval

- Filtering by call failure types
- Addressing (IP, OPC/DPC, and endpoint addresses)
- Call analysis

Using the Live Update Feature

Use Live Update to automatically query the data base for the call data corresponding to the most recent completed calls. This feature, shown in the following figure, allows you to set repeating query intervals to occur every 2, 5, 10, 30, or 60 seconds.



This feature will return call data from the 100 most recent calls.

Important: when searching for priority calls, such as '911' calls turn off the *Live Update* feature. *Live update's* 100 call maximum would exclude any high priority calls outside of that limit

Steps

To configure the Live Update feature:

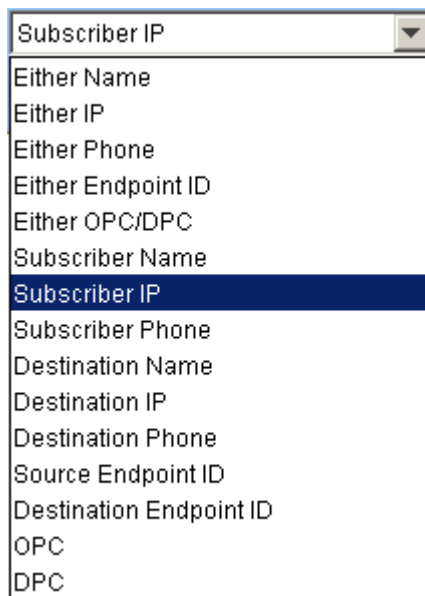
1. Click the down arrow in the *Live Update* interval drop down menu, and select the desired update interval.
2. Click the *Start Live Update* button to begin querying the most recent call data.

3. To disable Live Update, click the *Stop Live Update* button.

Steps

To search for an active call:

1. Select the *Active Calls* from the Diagnostics drop-down menu.
2. In the *ID Type* drop-down menu, shown in the following figure, select whether you are searching for the Originating Point Code (OPC), Destination Point Code (DPC), subscriber IP, phone number, or name, or destination IP, phone number, or name, or an IP or name from *either* subscriber or destination.

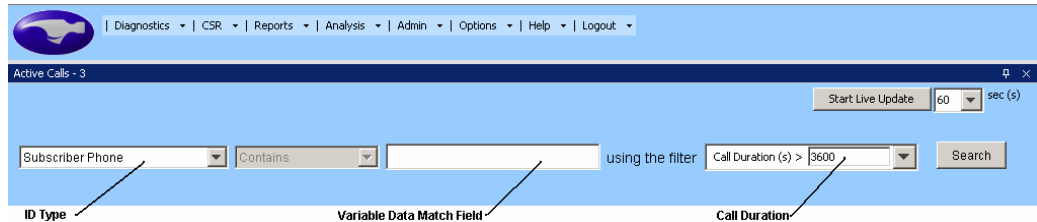


3. In the *Variable Data Match* field, enter an IP address or telephone number, or name, if possible.
4. To search for unusually long calls, or calls of known or approximate time length, select *Call Duration* from the filter drop-down menu, and

enter a call duration greater than the minimum seconds you are seeking.

For example, if you are trying to locate a call that has lasted longer than 1 hour, enter 3600 (the number of seconds in an hour) in the *Call Duration* value field, as shown below.

5. Click *Search*.



The screenshot shows the 'Active Calls - 3' window. At the top, there is a navigation bar with links: Diagnostics, CSR, Reports, Analysis, Admin, Options, Help, and Logout. Below this, there is a 'Start Live Update' button and a dropdown menu set to '60 sec (s)'. The main search area contains a dropdown menu for 'Subscriber Phone', a 'Contains' dropdown, an empty text input field, and a filter dropdown set to 'Call Duration (s) > 3600'. A 'Search' button is on the right. Three labels with arrows point to specific fields: 'ID Type' points to the 'Subscriber Phone' dropdown, 'Variable Data Match Field' points to the 'Contains' dropdown, and 'Call Duration' points to the filter dropdown.

Closing Calls in the Active Calls Window

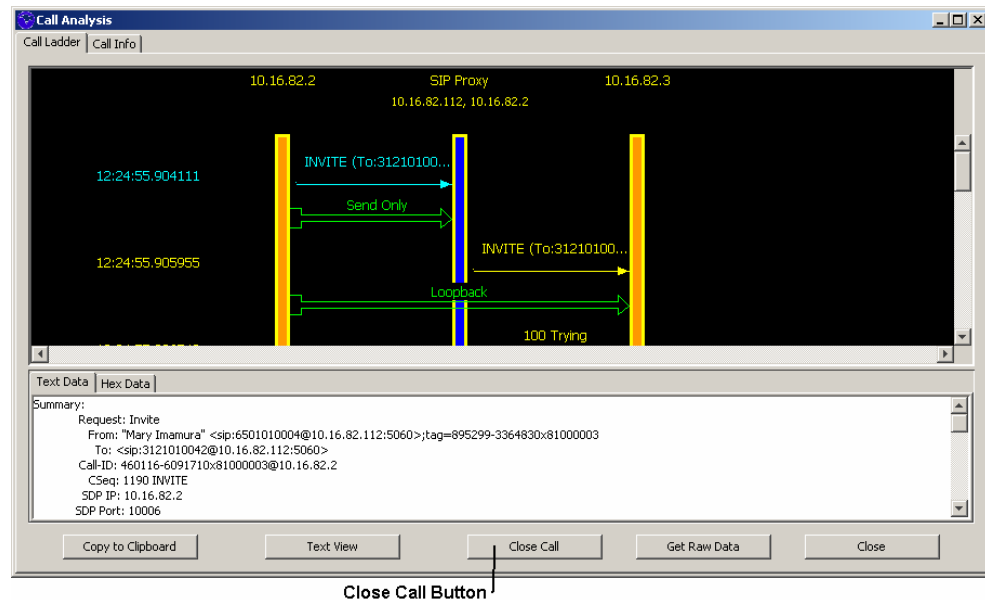
When the *Call Analysis* function is used for calls displayed in the *Active Calls* window, you may occasionally identify calls that have remained open due to signaling or packet handling errors. When this occurs it may be necessary to close the call using the *Close Call* button at the bottom of the *Call Analysis* window.

Steps

To close a call on the Active Calls page:

1. On the Diagnostics / Active Calls page, double-click the call you wish to close.

The Call Analysis window will display, as shown in the following figure.

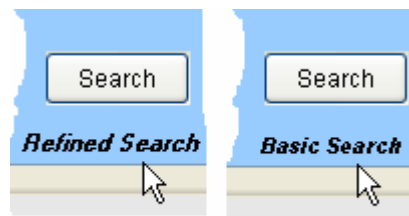


2. Click the *Close Call* Button. Once this has been done the call may still display in the *Active Calls* window. To clear the number from the *Active Calls* window after it has been closed, click the *Search* button a second time.

Searching for Completed Calls Using Basic and Refined Search Modes

The *Diagnostic/Completed Calls* page includes two search modes; *basic* and *refined*, accessed by clicking the *Basic / Refined Search* link located at the lower right side of the search window, directly beneath the *Search* button, as shown in the following figure.

Click this link to toggle back and forth between the two search modes.



Click Link

Use the *basic* search mode, shown below, when only subscriber or destination search parameters are needed to capture the required call data.

Use the *refined* search mode, shown in the following figure, when you are trying to:

- isolate subscriber IDs matching common search criteria.
- capture data from calls that occurred within a specific time interval
- list results in ascending or descending chronological order.
- identify tagged call data.

The screenshot shows a web application window titled "Completed Calls - 1". It contains several search and filter controls:

- Subscriber Phone**: A dropdown menu.
- Begins With**: A dropdown menu.
- Show results within the last**: A text input with "1" and a dropdown for "Hour(s)".
- Start**: A date/time picker showing "01/31/06 05:28 PM".
- End**: A date/time picker showing "01/31/06 06:28 F".
- Sorting**: Two radio buttons, "Newest First" (selected) and "Oldest First".
- Search Mode**: A dropdown menu set to "- No Filter -" with a "Search" button.
- Tag**: A dropdown menu set to "- None -" with a "Basic Search" label.
- Live Update**: A "Start Live Update" button and a "5 sec (s)" interval.

Annotations with arrows point to the "Start / End Date / Time Select" label and the "Ascending / Descending Sort Order" label.

Steps

When the Diagnostic window is first opened, the default search mode is *basic*.

Selecting a Search Mode

To select a refined search:

1. Click the *Refined Search* link shown in the figure, above.
2. Click the *Basic Search* to return to the default *Basic* search mode.

Performing a Basic or Refined Search

To perform a basic search:

1. In the Search For text box, enter a subscriber or destination telephone number, IP address, or user name.
2. In the Subscriber / Destination drop-down menu, select Subscriber or Destination.
3. To further refine your basic search, add an applicable filter from the Filter drop-down menu, or, if no filters fit your query requirements, leave the menu set to No Filter.

To perform a refined search

1. In basic search mode, click the *Refine Search* link to display the Refine Search dialog.
2. In the Subscriber / Destination / IP Address / Phone number drop-down menu, select a subscriber or destination phone number, IP address, or either.

If you are searching for a known ID (Tel number or IP Address), click the *Exact Match* box to locate information pertinent to the ID you have entered in the *Search for* field.

Note: By default, if a partial ID is entered, any IDs containing the value entered will be returned.

If you are searching by subscriber name *Joe Smith*, for example, make sure to place quotation mark immediately before *Joe*, and immediately after *Smith*, or the data returned will include all subscribers with a first name of *Joe*, **or** a last name of *Smith*.

3. To further refine your basic search, add an applicable filter from the Filter drop-down

menu, or, if no filters fit your query requirements, leave the menu set to *No Filter*.

4. In the *Search for* field, enter a partial Subscriber/Destination name, telephone number, or IP address, if you know a partial ID value, but not the exact or entire ID.

For example, you may only know a portion of the telephone number or IP Address. Using these search parameters, the query will return all the data matching your partial definition.

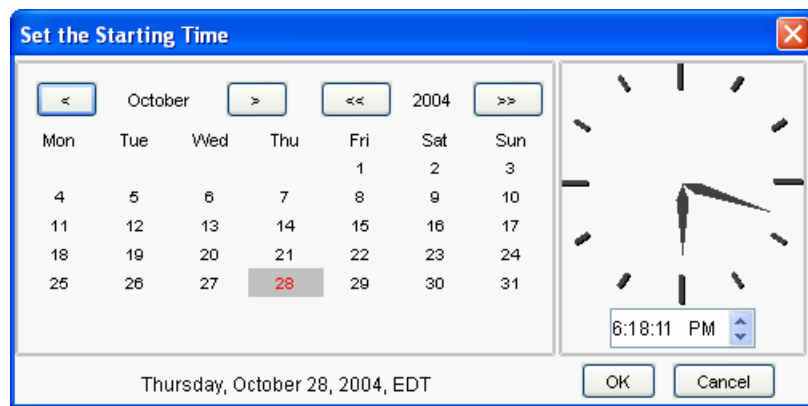
Note: If you are searching for call data based on a partial Subscriber/Destination ID, do not check the *Exact Match* box.

5. In the *Show Results Within The Last* drop-down menu, select the applicable time interval for your search.
6. Click *Newest First* or *Oldest First* to list your search results in ascending or descending chronological order.
7. When you have completed defining your query criteria, click *Search*.

Searching for Calls by Time Interval

To search for a call by time interval:

1. Select the Refined Search dialog.
2. In the *within the last* drop-down menu, select a predefined time interval from the drop-down menu, or select *User Defined* from the menu.
3. At the *Start* time box, click the clock Icon to open the *Starting Time* date/time dialog, as shown in the following figure.



4. Click the year and month arrow buttons to set the correct date.
5. Click the numeric day value to the correct day. Note that the current day appears in red text.
6. Click the up or down arrows beneath the clock display, or click and drag the hour and minutes hands to a set the correct time, and click *OK*.
7. At the End time box, repeat steps 3 through 6 above for the interval end time, and click *OK*.
8. Click *Search*.

Searching for Calls by IP Address or Phone Number

To search for calls by subscriber or destination IP address or phone number:

1. Select the Basic or Refined Search dialog.
2. In the *Source / Destination Type* field, select the type of source or destination ID from the drop down menu.

The options are:

- Either Name
- Either IP
- Either Phone

- Either Endpoint ID
 - Either OPC/DPC
 - Subscriber Name
 - Subscriber IP
 - Subscriber Phone
 - Destination Name
 - Destination IP
 - Destination Phone
 - Source Endpoint ID
 - Destination Endpoint ID
 - OPC
 - DPC
3. In the *Search Data* field, enter any specific identifier data you may have.
 4. Click the *Search* button

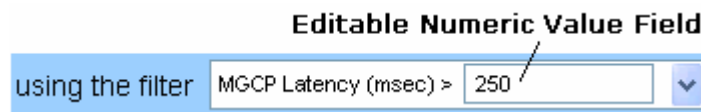
Searching for Calls Using Filters

To search for calls using a filter:

1. In either the Basic or Refined search, select a filter from the *filter* drop-down menu. The filter options include:
 - Bytes Recv > 50
 - Failed Calls
 - Connect Failed
 - Disconnect Failed
 - Call Duration (S) >250
 - Connect Latency (ms) >250
 - Disconnect Time (ms) >250
 - Error Code
 - Failure Reason

- Protocol Search
- Post Dial Delay (ms) >250
- Jitter (125us)>80
- Pkts Lost >50
- % Packets Lost >50
- MOS <3.0
- Packets Duped >50
- Packets Out of Sequence >50
- Packets Recv >50
- R Factor <70
- MGCP Jitter (msec) >10
- MGCP Packets Lost >50
- MGCP % Packets Lost >50
- MGCP Latency (msec) >250
- MGCP Bytes Recv.>50
- MGCP Packets Recv >50

The numeric values of those filters that contain numeric values, can be edited, once selected (see the following figure).

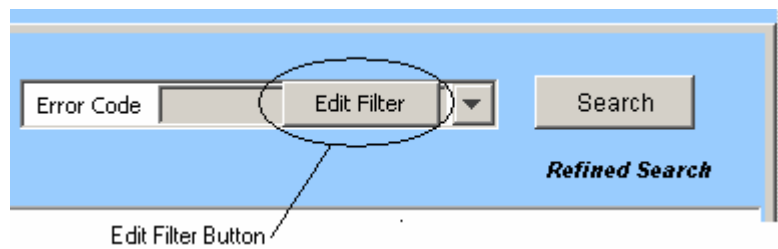


2. Select the time interval within which your search will occur.
3. Click the Search button.

Searching for Calls by Error Codes

To search for calls by Error Code:

1. In *refined search* mode, in the filter drop-down menu, select *Error Code*. The *Edit Filter* button will display when you mouse-over the text field, as shown in the following figure.



2. Click the Edit Filter button to display the *Error Codes* dialog, shown in the following figure.
3. Select the desired protocol tab to display the error codes for that protocol, and click the check boxes next to one or more error codes by which your call data search will be filtered.

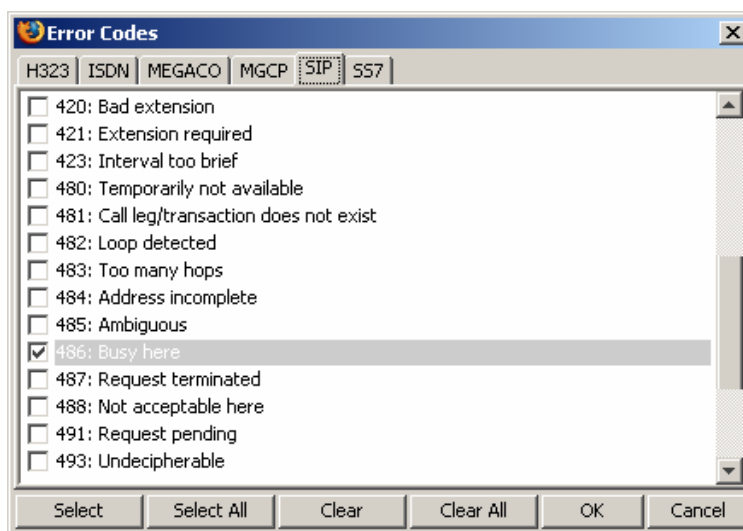
Note: Only tabs for protocols that are licensed for your Hammer XMS will appear.

To select all the errors within a protocol, click *Select* at the bottom of the dialog.

To select all the errors within *all* the supported protocols, click *Select All*.

Click *Clear* to clear all the error selections in any single protocol.

Click *Clear All* to clear all the error selections in *all* protocols.



As shown in this figure, SIP error #486 has been selected. Note that the selected error is now highlighted.

The resulting search will display those calls containing error codes for the protocols you have selected. If you select errors within more than one protocol, a listing of any calls containing ANY of the selected errors will be returned.

(refer to *Appendix A* of this document for a listing of applicable error codes).

4. Click *OK* to save the selected errors. Click *Cancel* to cancel your changes.
5. Set the time interval for your search query.
6. Click the *Search* button.

As shown in the following figure, the search returned a call containing a listing of the selected error, based on the search parameters.

The screenshot shows the 'Completed Calls - 3' window in the Hammer XMS Diagnostics page. The interface includes a navigation bar with links like Diagnostics, CSR, Reports, Analysis, Admin, Options, Help, and Logout. Below the navigation bar, there are search filters: 'Subscriber Phone', 'Begins With', 'using the filter', 'Error Code' (set to SIP:486), and a 'Search' button. There are also options to 'Show results within the last' (set to '- User Defined -') and 'and the Tag' (set to '- None -'). The 'Show results within the last' section has radio buttons for 'Newest First' (selected) and 'Oldest First'. The 'Start' and 'End' time ranges are set to '02/07/06 02:57 PM' and '02/08/06 03:57 F' respectively. A 'Start Live Update' button is also present. Below the filters is a table with the following data:

Time	Subscriber Name	Source Address	Subscriber Phone	Destination Name	Destination Address	Destination Phone	Duration	Fail Reason
Wed Feb 08 14:50:47 EST 2006	NT CLINIC GILL M	70.184.192.38	5403445427		68.109.201.40	5408532591	00:00:04	SIP: Busy here;

Searching for Calls Using Tags

If your Hammer XMS system includes the Tag option license, and your system administrator has created tags for the system, you can search for tagged call data using the Diagnostics page search functions or the Analysis page report formats.

Steps

To search for tagged calls:

1. On the *Diagnostics* page, in *Refined Search* mode, select a tag in the *and the tag* drop-down menu, along with any additional search parameters.
2. Click the *Search* button. The Hammer XMS will return all calls that contain the selected tag.

In the following figure, a tag called “IP Denver” had been selected, and a search has returned all those calls consistent with that tag.

using the filter

and the Tag **Basic Search**

Origin IP	Destination Phone	Duration	Tags
168.0.17	1234567890	5	IP Denver
168.0.17	5678901234	5	IP Denver
168.0.17	2345678901	5	3 Tags
168.0.17	1122345678	5	IP Denver
168.0.17	9012345678	5	IP Denver
168.0.17	4567890123	5	IP Denver
168.0.17	3456789012	5	IP Denver
168.0.17	6678901234	5	IP Denver
168.0.17	7890123456	5	IP Denver
168.0.17	1324354657	5	IP Denver

Note that the third row down, in the Tags column displays “3 Tags”. This indicates that the call also contains two additional tags.

using the filter

and the Tag **Basic Search**

Origin IP	Destination Phone	Duration	Tags
168.0.17	1234567890	5	IP Denver
168.0.17	5678901234	5	IP Denver
168.0.17	2345678901	5	IP Denver
168.0.17	1122345678	5	IP Atlanta
168.0.17	9012345678	5	IP NY
168.0.17	4567890123	5	IP Denver
168.0.17	3456789012	5	IP Denver
168.0.17	6678901234	5	IP Denver
168.0.17	7890123456	5	IP Denver
168.0.17	1324354657	5	IP Denver

3. To identify which other tags apply to that call, click in the *3 Tags* cell. A drop-down menu, shown in the previous figure, will display those additional tags.

Using Advanced Column Features

The advanced column sorting features allow you to:

- Sort and sub-sort columns.
- Set the default in columns for ascending or descending order.

Using the Column Sort / Sub-sort Feature

Use the column sort / sub-sort feature to create ordered lists and list subsets.

For example; you can list all calls associated with a particular IP address as the primary listing criterion.

Within those calls, you may want to list them in chronological order of date. This would become your first subset.

Finally, you could create a list of call length in descending order, for each date displayed. This would be a second subset.

Steps

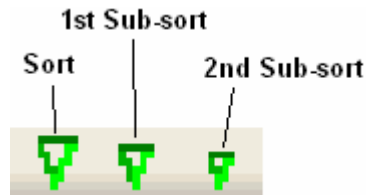
To sort / sub-sort your call lists:

1. In the Call Diagnostics window, perform a search as described above.
2. Select a column to sort by clicking the column header, as shown in the following figure.
3. To sub-sort, hold down the <Ctrl> key, and click the column header of a second column you want to sub-sort.

4. To add a second sub-sort, hold down the <Ctrl> key, and click the column header of a third column you want to sub-sort.

By default, call analyses are sorted in descending order, as indicated by green downward pointing arrows that appear in the header cells you select.

The arrows decrease in size from initial sort, 1st sub-sort, and 2nd sub-sort. See the following figure.



Example: To sort by subscriber name, sub-sort within those names by call duration, and then sub-sort within call duration by the time the calls occurred:

1. Perform your search as described above.
2. Click the header cell of the *Subscriber Name* column. A green arrow will appear.
3. Hold down the <Ctrl> key, and click the header cell of the Duration column. A second, smaller arrow will appear in that header cell.
4. Hold down the <Ctrl> key, and click the header cell of the Time column. A third arrow (smallest of the three) will appear.

To change the order from descending to ascending, click a second time in the header cell

Viewing Call Details

Use the features provided by the *Call Details* view, shown in the following figure, to view call states or identify problems associated with individual calls.

The screenshot shows the 'Call Details' window with two tabs: 'Call Ladder' and 'Call Info'. The 'Call Info' tab is active, displaying call data metrics and call legs.

Call Data

Start Time:	Fri Mar 04 11:40:19 EST 2005	Call Duration (s):	14
End Time:	Fri Mar 04 11:40:34 EST 2005	Post Dial Delay (ms):	1889
Subscriber:		Connect Latency (ms):	0
IP:	24.168.151.226	Disc. Time (ms):	0
Phone #:	2015961625	Tags:	Global-gold.dca
Display Name:	Hrand Topjian		
Destination:			
IP:	162.97.226.252		
Phone #:	19739021122		
Display Name:			
Failure Code:	0	Reason:	SIP: - Cancel; SIP: - Cancel; Cancel; Cancel

Call Legs

SIP	Media Src Port:	19492
SIP	Media Dst IP:	147.135.0.128
SIP	Media Dst Port:	12350
SIP	Payload:	G.711
SIP	R Factor:	92
OutBound Media	MOS:	4.17969
InBound Media	Packets Received:	626
OutBound Media	Octets Received:	136468
	Packets Lost:	0
	Packets Duped:	0
	Packets OOS:	0
	Jitter Avg. (ms):	0
	Jitter Min. (ms):	0
	Jitter Max. (ms):	0

Buttons: Text View, Close

The Call Details window provides two tabs – *Call Ladder*, and *Call Info*. The *Call Info* tab provides the following capabilities:

- Displays general Call data metrics, such as *Start Time*, *End time*, *Duration*, *Post Dial Delay* and *Call Latency*.
- Displays Call legs by protocol type.
- Displays Call failure codes, and their cause.

- Provides a text version of the data that can be copied to your computers clipboard, printed, or emailed.

The *Call Ladder* tab provides:

- A graphical depiction of the call ladder and legs.
- Data from individual calls in hex and ASCII formats.
- Buttons to copy either hex or ASCII versions of the ladder diagram and message data to your computers clipboard for pasting into other applications.

Steps

To view call details:

1. In the Diagnostic window click to highlight and select the call you want to analyze, and click the *Show Detail* Button.

Alternately, double-click the number to select and launch the call analysis.

Completed Calls - 1

Start Live Update 5 sec(s)

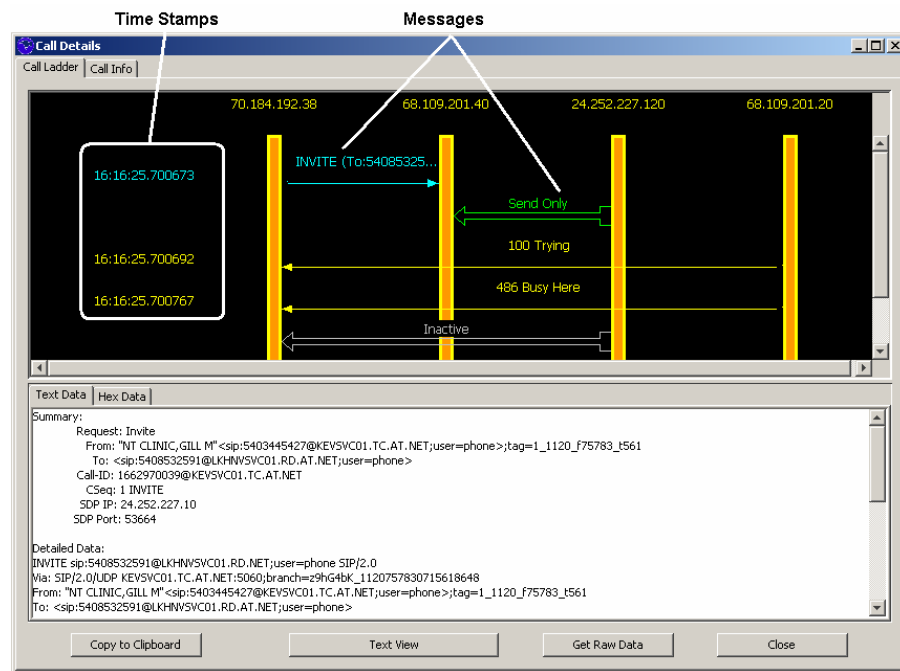
Subscriber Phone Begins With using the filter Error Code SIP:486 Search

Last Hour Today Last 2 Days This Week

Time	Subscriber Name	Source	Subscriber Phone	Destination Name	Destination Phone	Duration	Fail Reason
Sun Jan 22 16:16:25 EST 2006		70.191.1.1		24.246.8701067		00:00:01	
Sun Jan 22 16:16:25 EST 2006		172.18.1.1		172.18.1.1		00:00:00	
Sun Jan 22 16:16:25 EST 2006		172.18.1.1		172.18.1.1		00:00:01	
Sun Jan 22 16:16:25 EST 2006		172.18.1.1		172.18.1.1		00:00:00	
Sun Jan 22 16:16:25 EST 2006		64.74.441447506506		198.313954011532142670		00:00:00	
Sun Jan 22 16:16:25 EST 2006		64.74.441447506506		198.313954011532142670		00:00:00	
Sun Jan 22 16:16:25 EST 2006		172.18.1.1		172.18.1.1		00:00:00	
Sun Jan 22 16:16:25 EST 2006	NT CLINICGILL M	70.184.5403445427		68.1055408532591		00:00:04	SIP: Busy here;
Sun Jan 22 16:16:25 EST 2006		64.74.441447506506		198.313954011532142670		00:00:00	
Sun Jan 22 16:16:25 EST 2006		64.74.441447506506		198.313954011532142670		00:00:00	
Sun Jan 22 16:16:25 EST 2006		70.184.1.1		24.246.8701067		00:00:00	
Sun Jan 22 16:16:25 EST 2006		70.191.1.1		24.246.8701067		00:00:00	
Sun Jan 22 16:16:25 EST 2006		172.18.1.1		172.18.1.1		00:00:00	

Calls found: 6000 Show Detail

The Call Details window opens, as shown in the following figure.



2. Click any leg of the ladder to see the text version of the leg data displayed in the Text Data tab window.
3. Click the *Hex Data* tab to view the information in hex format, as shown in the following figure.


```

Text Data  Hex Data
[1, MGCP->NTFY, 121 bytes, 10.33.40.160, 172.16.251.8, 2427, 2427]

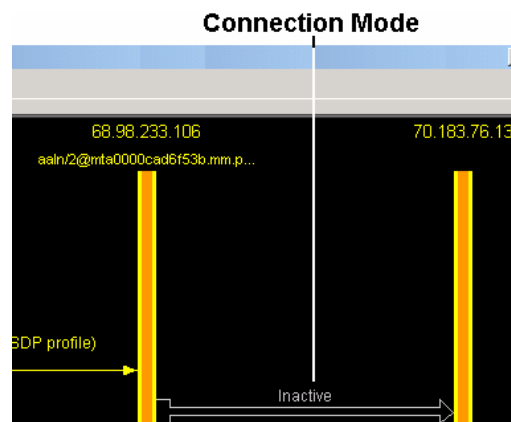
0000 00 01 AF 02 C8 D9 00 03 4B 5D F6 01 08 00 45 00 .....K]....E.
0016 00 6B A5 FF 00 00 3C 11 FE A8 0A 21 28 A0 AC 10 .k....<....!(...
0032 FB 08 09 7B 09 7B 00 57 45 4B 4E 54 46 59 20 34 ...{(.WEKNTFY 4
0048 34 32 20 61 61 6C 6E 2F 32 40 73 74 70 74 77 69 42 aaln/2@stptwi
0064 38 38 36 36 34 30 30 31 2E 63 68 61 72 74 65 72 88664001.voip.net
0080 76 6F 69 70 2E 6E 65 74 20 4D 47 43 50 20 31 2E MGCP 1.0 NCS
0096 30 20 4E 43 53 20 31 2E 30 0A 58 3A 20 31 35 31 1.0.X: 151 18.0: hd.
0112 31 38 0A 4F 3A 20 68 64 0A

```

a?

Understanding Connection Mode

The Connection Mode indicator, shown in the following figure, displays messaging within displayed call ladders to indicate signaling or media packet stream status.



Possible messages include:

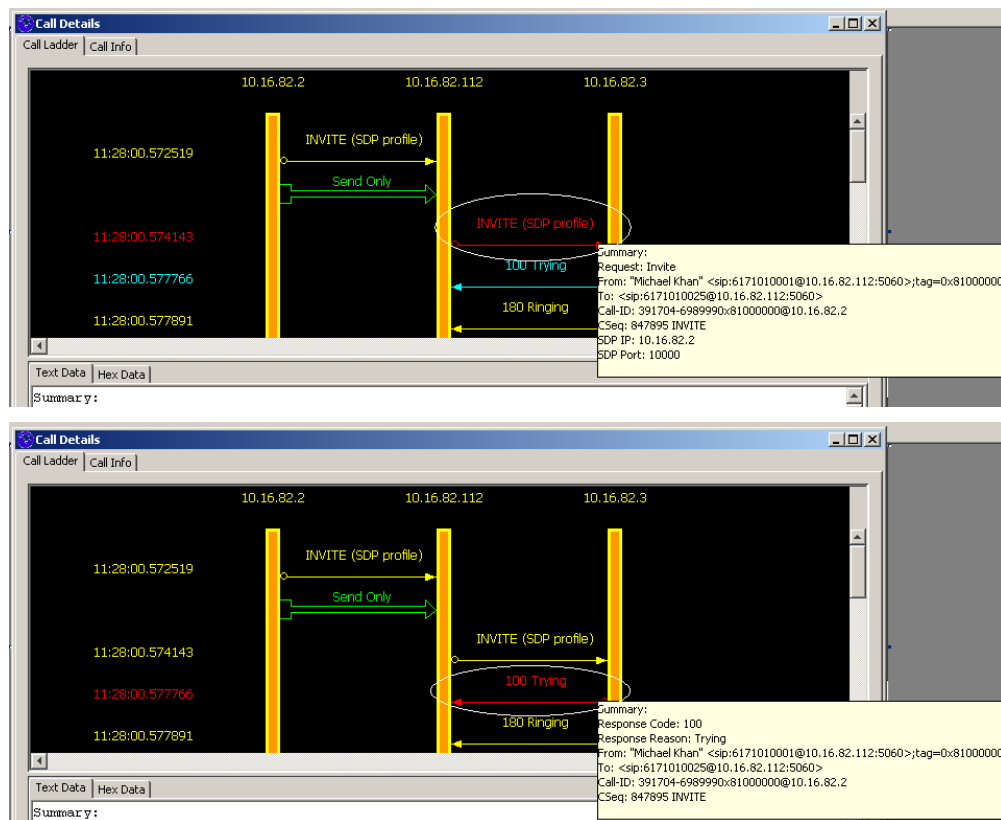
- Inactive
- Receive Only
- Send Only
- Send Receive

Using the Call Leg Pop-up Summary

An additional feature in HXMS Release 1.5 is the ability to see access leg-specific data when mousing over a call leg.

In the first of the two following figures, the user has moused over the *Invite* message between points 10.16.82.112 and 10.16.82.3 (circled in white). Note that the specified leg turns red. Data for both the call, and the message are provided in the summary pop-up.

In the second figure, the user has moused over the *Trying* message between the same two points. And again, a summary popup appears.



Using *Protocol, Legend, Time Stamp, Display Options*

Use the Show Protocol Indicators feature to display protocol specific markers on your call legs to make them visually identifiable.

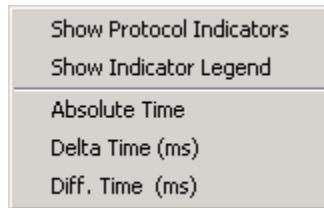
The options include:

- Protocol Indicators
- Protocol Indicator Legend
- Time Stamp Type
- Summary Info Pop-up

Steps

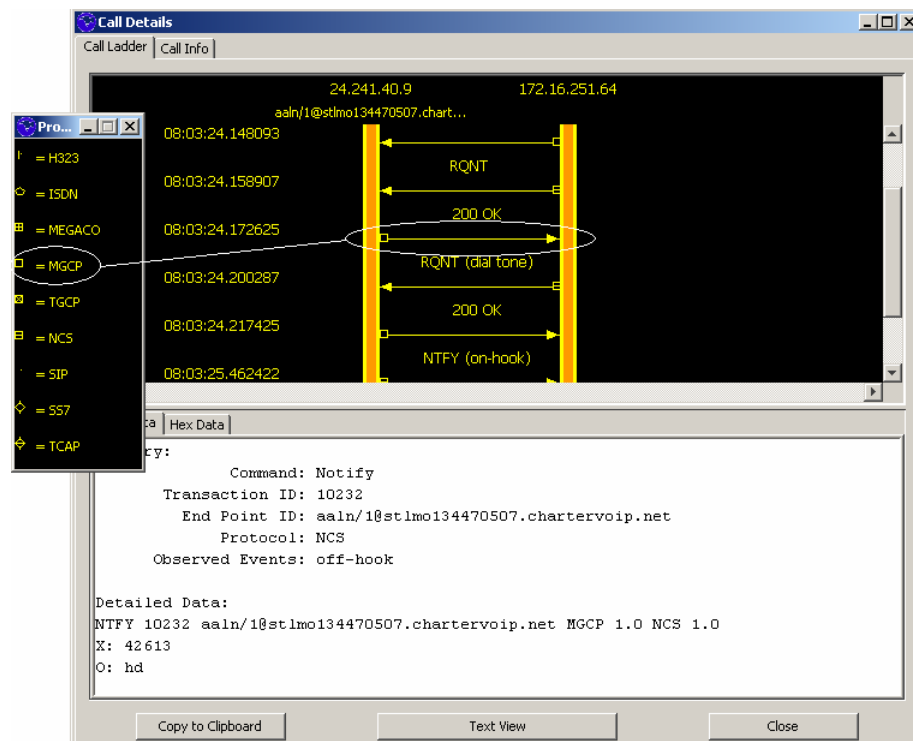
To use the protocol / legend feature:

1. Right-click any call leg (or its time stamp at the left side of the ladder window) to display the Legend / Time menu, shown in the following figure.



2. Click the *Show Protocol Indicators* option to display a protocol symbol on each call leg.
3. Click the *Show Indicator Legend* option to display the protocol symbol set, shown at the left in the following figure.

The symbol will show on the origination end of any call leg, as shown.



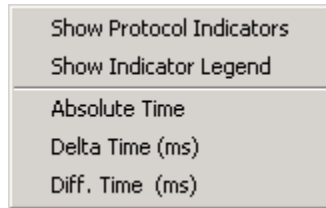
The timestamp value provided at the left side of the ladder view can be set to display three types of timestamp data:

- Absolute Timestamp (Default) – The timestamp in hours, minutes, seconds, and microseconds of the call frame.
- Delta Timestamp – The time difference between the first call frame and the current call frame.
- Difference Timestamp – The time difference between the previous call frame and the current call frame.

Steps

To select the type of timestamp displayed with call ladder events:

1. Right-click the timestamp to the left of the ladder in the call ladder view. The following menu will appear (refer to the ladder figure on page 34 to see the menu within the ladder view).



2. Select the timestamp value you desire.

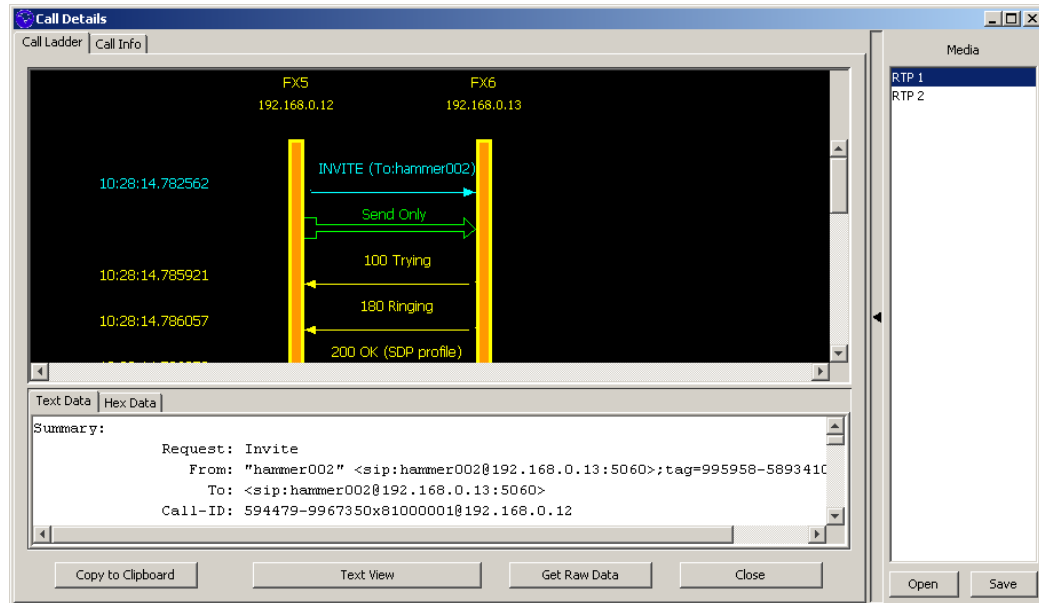
Recording, Playing, and Saving RTP

Hammer XMS, Release 1.5 provides the ability to record, play, and save RTP streams.

Steps

To capture RTP audio data:

1. In the *Diagnostics* Search window, perform a search the call or calls you wish to analyze.
2. From the search results, double-click any call from which to extract RTP. The *Call Details* window will display, as shown in the following figure. Note the *Media* pane at the left side of the window. This will display any RTP streams contained in the selected call.
3. In the *Media* pane, click to highlight the desired RTP stream.

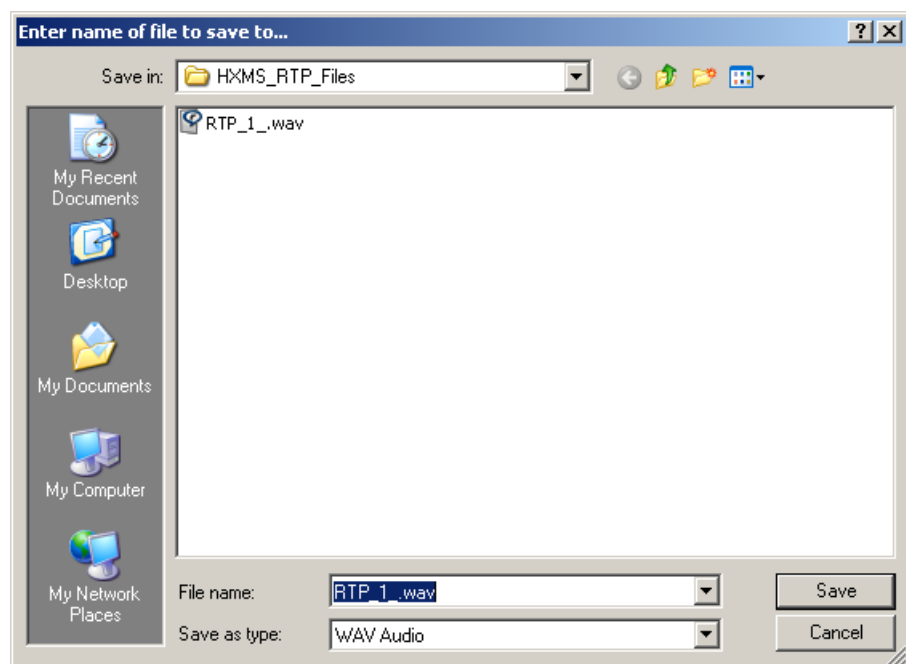
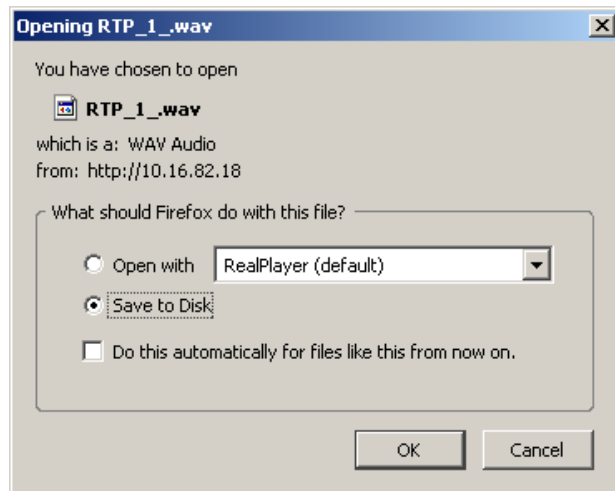


4. At the bottom of the Media pane, click *Open* to play the data, using your default audio application such as QuickTime. Or click *Save* to save the file to the desired location.

The *Opening* dialog will appear, as shown in the following figure.

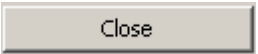
5. At the bottom of the Opening dialog, click *OK* to open a file manager dialog.

In the example provided in the bottom figure, below, the user has created the folder *HXMS_RTP_Files* into which all RTP files will be saved.



Using other Call Details Features

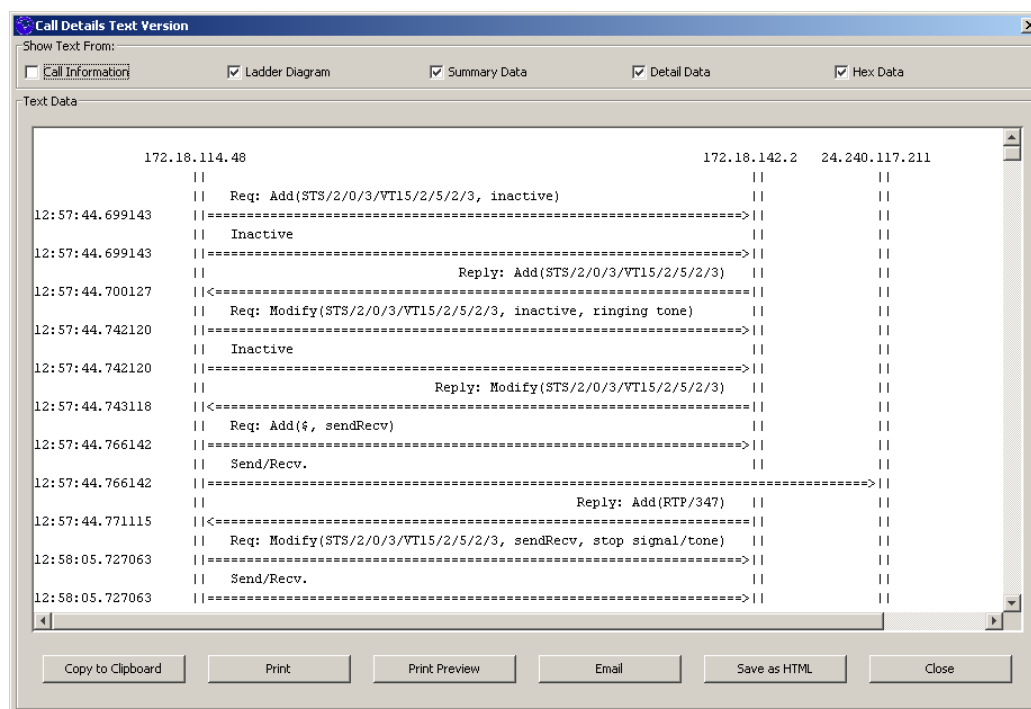
At the bottom of the Call Details window are the following buttons:

Button	Function
	Use the <i>Copy to Clipboard</i> button to copy the call leg data to your computers clipboard for exporting to other applications, such as spreadsheet or publishing tools. If you highlight a portion of the displayed text, only the highlighted text will be stored to clipboard. If you do not highlight specific text, all of the text displayed will be copied to the clipboard.
	Use the <i>Text View</i> button to display the text version of the entire call.
	Use the <i>Close</i> button to close the window.

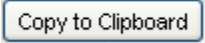
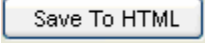
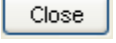
Clicking the Text Version button displays the text version of the call analysis, shown in the following figure. This feature is particularly useful for exporting comprehensive call data into another application.

Use the check boxes at the top of the dialog to select the type of data that you want to see displayed.

Use the buttons at the bottom of the dialog for the functions described in the following table.



The following table describes the buttons located at the bottom of the Text Version window.

Button	Function
	Use the <i>Copy to Clipboard</i> button to copy the entire call analysis text to another application. To copy only a portion of the text, highlight the desired portion before clicking the button.
	Use the <i>Print</i> button to send the text to a print queue.
	Use <i>Print Preview</i> to format the print version prior to sending it to the print queue.
	Use the <i>Email</i> button to open the mail client to which your browser defaults.
	Use the <i>Save To HTML</i> button to create an HTML version of the Call Detail data.
	Use the Close button to close the window.

Using Text View Email

Hammer XMS allows the user to email text versions of the *Ladder Diagram*, and the *Call Info* tab. When the *Email* button at the bottom of the Text View window is clicked, all of the text in the dialog is selected and copied to the clipboard.

The Hammer XMS software then opens the default email client of the browser, and notifies the user that the data has been copied to the clipboard for their convenience.

Important! Make sure to disable any pop-up blocking software, prior to using the text view email feature.

The message also directs the user to remove the instructional message and replace it with the text copied to the clipboard.

Prior to using the Email function, Empirix recommends that you verify that your browser default email client settings are correct. If it is not, contact your system administrator for assistance.

Steps

To verify your browser default email client settings for *Internet Explorer*:

1. Click the Tools menu, at the top of the browser window, and select Internet Options, and then select the Programs tab.

The dialog then displays a list of actions and programs that are selected as the defaults for those actions.

2. Click on the drop down list for email to select your default application.

To verify your browser default email client settings for *Netscape* or *Mozilla* browsers:

1. Click on the email button. Netscape and Mozilla both have their own default email clients.
2. If you do not have an account with Netscape or Mozilla a dialog will appear that will help you create one. Once an account is created then you will be able to copy and paste the text in an email. Contact your system administrator for further information.

Reviewing Call Information

The *Call Info* tab provides a variety of information about analyzed calls, including:

- Addressing information (source and destination).
- Time intervals associated with the call from initial messaging to disconnect.
- Delay, Jitter, and Latency.
- Packet metrics.

The following table identifies how each of these metrics are defined, for each of the supported protocols.

Industry Name	Call Duration	Network Response	Post Dial Delay	Answer Delay	Release Response
HXMS Name	Call Length	T1	Post Dial Delay T2	Connect latency T3	Disconnect Time T4
Description	Total Length of call	First Response From Network	Last Digit - First Ring	Total Connect Time	Time to Disconnect
SIP	1 st Invite to 200 bye response	1 st Invite to 100	1 st Invite to 180 response	1 st Invite to ACK	Bye to 200 bye response
SS7	IAM to RLC	IAM to ACM		IAM to ANM	REL to RLC
MGCP	NTFY or CRCX (hd) to 250 response to DLCX	CRCX - 200 (CRCX transaction)	NTFY (digits) or CRCX to 200 response to CRCX	NTFY (hd) or CRCX to 200 to MDCX	DLCX or NTFY (hu) to 250 response to DLCX
MEGACO	NTFY (al/of) or NTFY (dd/ce) or Request:Add TO last Reply: Subtract or ResponseAck (Subtract)	First Request:XXX TO First Reply:XXX	Notify (dd/ce) or Notify (dd/dx) or Request:Add TO Request:XXX w/signal (cg/rt, cg/bt, cg/ct, cg/sit, cg/wt, cg/prt, cg/cw, cg/cr) or Reply:Add	Notify (al/of) or Request:Add TO Request/Reply:XXX that provides both src/dst SDP	Notify (al/on) or first Request: Subtract TO last Reply:Subtract
ISDN	Setup TO Release Complete	Q.931 Setup TO Setup ACK or Call Proceeding or Alerting or Connect	Setup TO Alerting or Connect	Setup to Connect ACK or Connect	Disconnect or Release TO the Release Complete

As shown, call metrics such as Call Duration, Network Response, Post Dial Delay (PDD), Answer Delay, and Release Response vary somewhat between SIP, SS7, H.323 and MGCP protocols.

Steps

To view call info:

1. At the upper left corner of the Call Analysis window, click the *Call Info* tab to view detailed call information, as shown in the following figure.

Call Info Tab

Call Details

Call Data

Start Time:

Subscriber

IP:

Phone #:

Display Name:

Destination

IP:

Phone #:

Display Name:

Call Duration (s):

Post Dial Delay (ms):

Connect Latency (ms):

Disc. Time (ms):

Tags:

Failure Code: Reason:

Call Legs

SIP

OutBound Media

InBound Media

Start Time:

Src IP:

Src Phone #:

Src Display Name:

Dst IP:

Dst Phone #:

Dst Display Name:

Media Src IP:

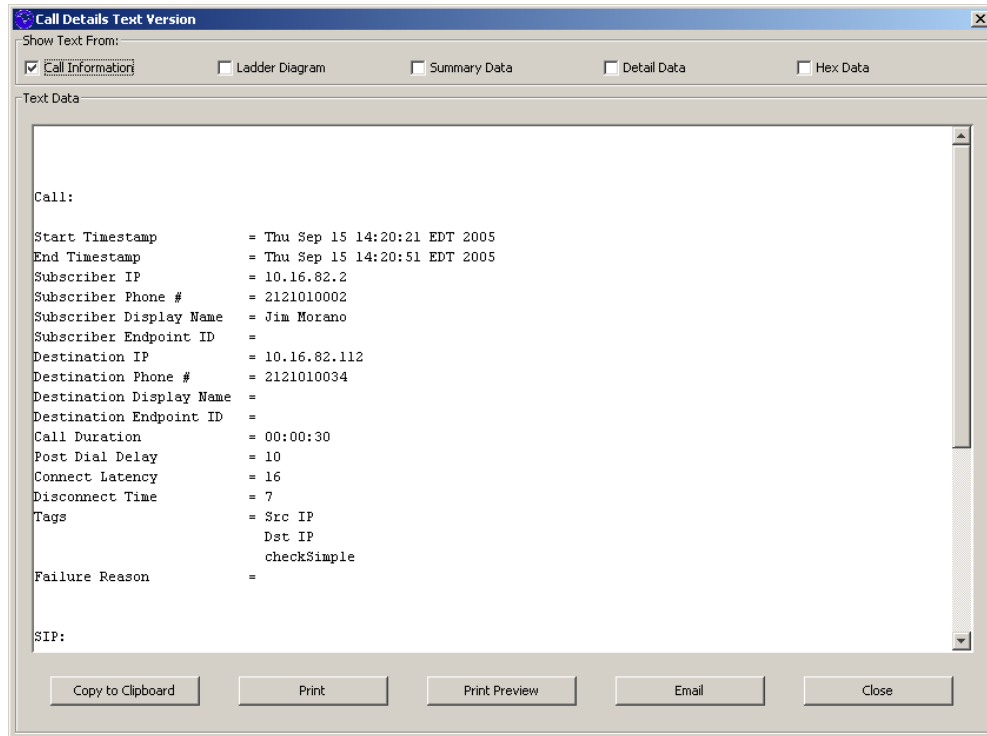
Media Src Port:

Media Dst IP:

Media Dst Port:

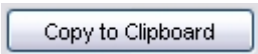
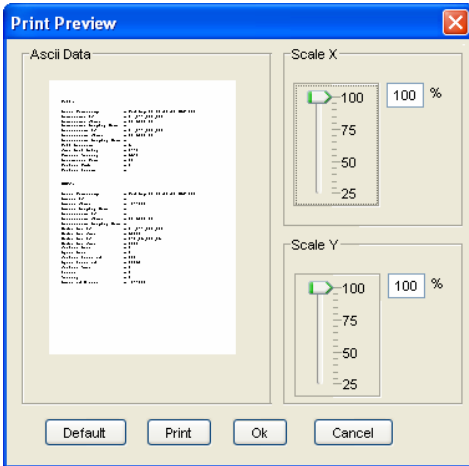
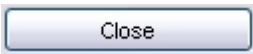
VLAN ID:

2. At the bottom of the window, click the *Text View* button to view the data in text format (see following figure).



Check any or all of the five check boxes at the top of the window to select the type of data displayed.

The Call Data Text Format page also includes five buttons at the bottom described in the following table:

Button	Function
	Use the Copy to Clipboard button to copy the text data to another application such as a spreadsheet or publishing program. If you highlight a portion of the displayed text, only the highlighted text will be stored to clipboard. If you do not highlight specific text, all of the text displayed will be copied to the clipboard.
	Use the Print button to send captured data to a print queue.
	Use the Print Preview function, shown in the following figure, to format call data text for printing, prior to sending it to a print queue. 
	Use the Email button to open your browsers default mail client.
	Use the Close button to close the Call Details window.

Note: The Print, Page Format, and Email functions described above are known to run on

MS-Windows OS, with the browsers listed in the following table. For all other operating system, mail client, and browser types, contact your administrator for assistance.

Browser	Version(s)
Internet Explorer	6.0 and newer
Mozilla	1.7 and newer
Netscape	7.1 and newer

Using the Analysis Feature

The Hammer XMS Analysis feature, shown in the following figure, provides users with value-added set of tools to interactively view and analyze active call data in the following report categories:

- Calls
- Errors
- Media
- Busy Call Hour Analysis.

When performing call data searches, its important to remember that the Hammer XMS *Analysis* and *Reports* pages calculates call time somewhat differently than the *Diagnostics* page (see following figure).

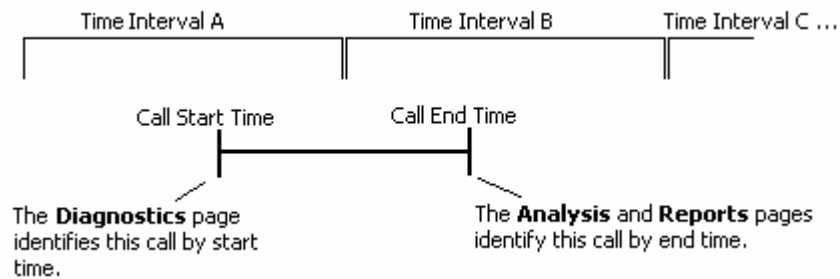
The information displayed in the *Analysis* and *Reports* pages is derived from completed calls that have been stored in the database. This information comes from data that is continually sampled and summarized as the signaling messages that accompany calls are processed.

This summarized data is written to the database at regular intervals, typically once per minute. Data associated with a given call – for example, *call*

duration and *connect latency*, is not available to the sampling process until the call completes.

For this reason, the summarized statistics represents a time interval that contains the ending time of the call, even though the call may have started minutes, or even hours, before. Thus, the *Analysis* and *Reports* pages identify calls according to their END times.

When *Show Calls* feature is launched from an *Analysis* page row (or distribution chart) for a particular analysis time period – typically to select specific calls for examination – the calls that are displayed are those that *ended* within the defined time interval. Thus, the actual *call starting time* may be from a time interval that is earlier than the interval displayed in the *Analysis* page view.



This means that the call overlapped the defined period, as shown in the following figure.

Right-clicking individual cells of Analysis reports provides one or more of the following functions, depending on that functions relevancy for the data contained in a particular column:

- Output comma-separated value (CSV) files for spreadsheet applications.
- Show call data in historical and distributed graph form.

- Generate call data tables.
- Display tagged call data.

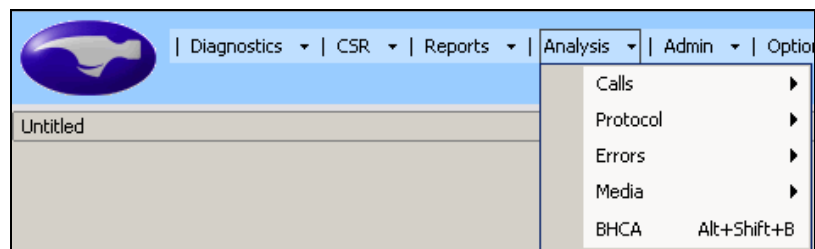
However, the *Diagnostic* page identifies calls according to their START times. This is done for two reasons:

- To be consistent with the active call diagnostic display (active calls have no end times).
- In most cases a subscriber seeking help for a calling problem will indicate the time when a call was placed (or attempted), rather than when the call ended.

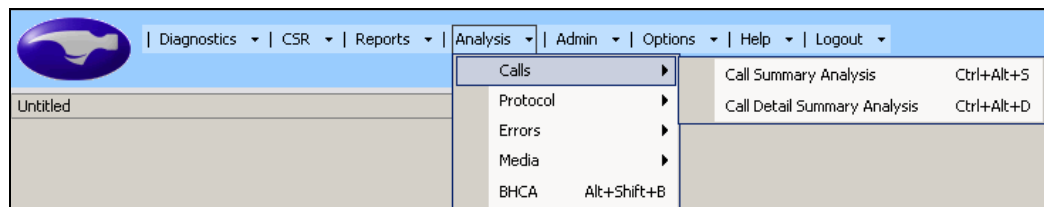
Steps

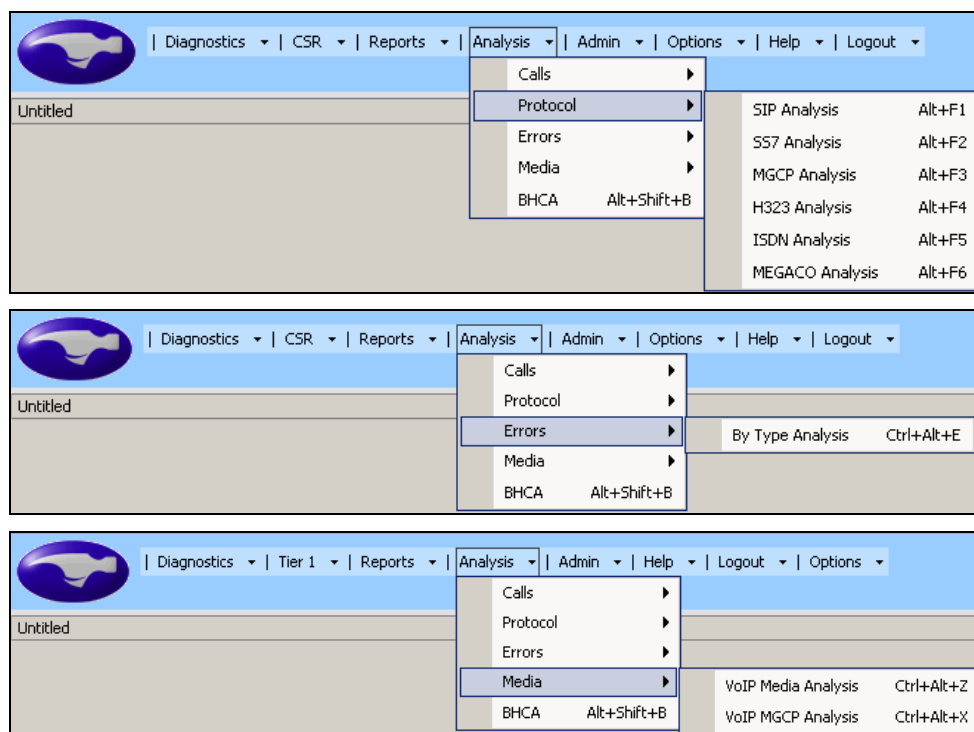
To use the Analysis feature:

1. At the top of the main HXMS page, click the *Analysis* link, as shown in the following figure.

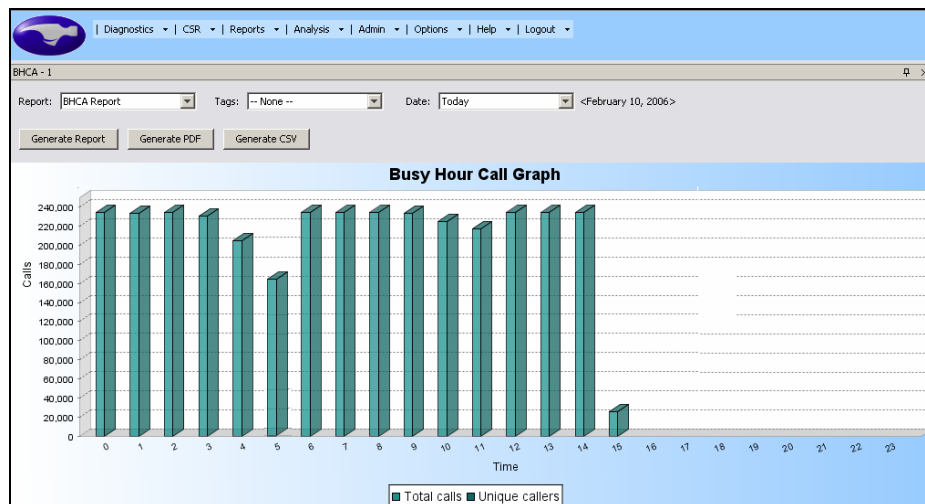


2. Mouse-over each of the first four menu selections to see the available analysis report options for that selection (see following figures).





3. Click the analysis you wish to view. An example BHCA Analysis Report is shown in the following figure.



4. At the bottom of the page, select a time interval or time range for the analysis. Clicking the *Pick A Range* link opens the dialog shown in the following figure.

The image shows a 'Date Range' dialog box with a blue title bar and a close button. It contains two calendar views. The top view is for January 2005, with the 1st highlighted. The bottom view is for February 2005, with the 1st highlighted. Both views have navigation buttons for months and years. At the bottom are 'OK' and 'Cancel' buttons.

Pick The Start Time:

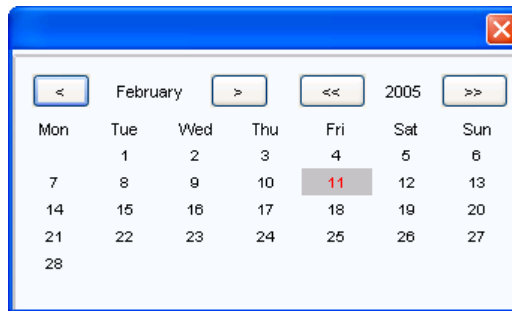
Mon	Tue	Wed	Thu	Fri	Sat	Sun
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

Pick The End Time:

Mon	Tue	Wed	Thu	Fri	Sat	Sun
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28						

OK Cancel

- Use the forward/backward arrows to select the month values. Use the forward/backward double arrows to select year values
- Click to select a start date from the upper half of the dialog, and click an end date from the lower half.
- Click OK when start and end dates have been selected. The Pick A Date dialog, shown in the next figure, is a simplified version of the Pick A Range dialog.
- Select the month, year, and day, as described above.
- Click OK.



Outputting data to a PDF file

You may also output the captured data to an Adobe PDF file, by clicking *Generate PDF*.

- Right-click any table cell to select options for the type of metric in that column. A menu of options will appear, as shown in the following figure. The available options will vary, depending on the type of information contained in the cell you are viewing.

Call Detail Summary- Analysis for Sep 12, 2005

Hour	Calls Attempted	ASR	Calls Completed	Calls Failed	% Calls Failed	Avg CPS	Avg Ca
Mon Sep 12 00:00:00 EDT 2005	283	96.4664	283	10	4	0	00:00:19
Mon Sep 12 01:00:00 EDT 2005	266	98.4962	266	4	2	0	00:00:19
Mon Sep 12 02:00:00 EDT 2005	268	97.3881	269	7	3	0	00:00:19
Mon Sep 12 03:00:00 EDT 2005	203	98.5222	203	3			00:00:20
Mon Sep 12 04:00:00 EDT 2005	160	99.3750	160	1			00:00:23
Mon Sep 12 05:00:00 EDT 2005	162	99.3827	162	1			00:00:22
Mon Sep 12 06:00:00 EDT 2005	179	99.4413	179	1			00:00:23
Mon Sep 12 07:00:00 EDT 2005	387	97.4160	385	10			00:00:21
Mon Sep 12 08:00:00 EDT 2005	527	95.2562	526	25			00:00:20
Mon Sep 12 09:00:00 EDT 2005	518	96.1390	519	20	4	0	00:00:20
Mon Sep 12 10:00:00 EDT 2005	518	95.7529	518	22	4	0	00:00:20
Mon Sep 12 11:00:00 EDT 2005	515	96.5049	516	18	3	0	00:00:20
Mon Sep 12 12:00:00 EDT 2005	556	96.0432	554	22	4	0	00:00:21
Mon Sep 12 13:00:00 EDT 2005	571	95.0963	571	28	5	0	00:00:21
Mon Sep 12 14:00:00 EDT 2005	585	92.8205	585	42	7	0	00:00:21
Mon Sep 12 15:00:00 EDT 2005	567	94.5326	567	31	5	0	00:00:22

- Save Table to a CSV File
- Show Calls
- History Graph
- Show by Type

[Today](#)

[Yesterday](#)

[Pick A Range](#)

[Pick A Day](#)

[Generate PDF](#)

A brief description of these options is provided in the following table. A more detailed look at the *Show Calls*, *History Graph*, and *Distribution Graph* functions follows the table.

Option	Description
Save Table to CSV File	Saves the selected data in an ASCII comma-separated value file to a location of your choosing.
Show Calls	Displays a list of calls for the interval represented by the row in which you right-clicked. This feature is available for all columns except <i>Calls per Second (CPS)</i>, <i>Answer Seize Ratio (ASR)</i>, and <i>% Calls Failed</i>.
History Graph	Displays a graph in increments of the interval of the row in which you've right-clicked. This feature is available in all columns. Example – if the row you've selected is a one hour interval, the resulting Historical Graph will show activity for each of the minutes comprising that hour.
Distribution Graph	Displays the distribution of captured call data within the interval selected. This feature is available for the Call Length, PDD, Connect Latency (CL), and Disconnect Time (DT) columns.
Show by Type	Displays a graph of call errors by protocol type and error ID.

Creating and Viewing CSV Files

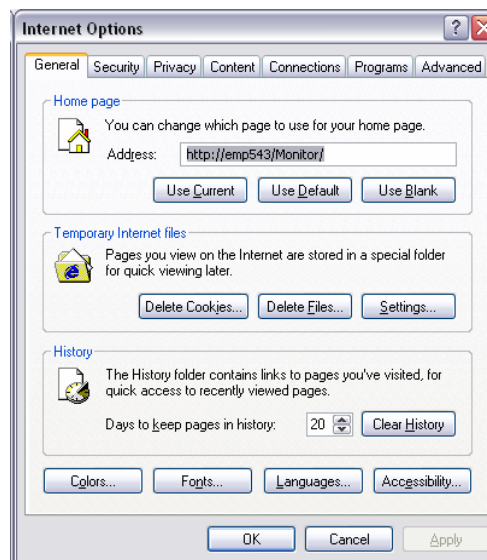
The CSV File function saves the selected data as an ASCII comma-separated value file, to a location of your choosing. This type of file is readily supported by common spreadsheet applications, such as Microsoft Excel.

Important: Internet Explorer requires that the Hammer XMS server be recognized as a *trusted site* in order to allow saving of CSV data files. If you are using Internet Explorer, follow these steps to configure Internet Explorer to allow you to save CSV data files.

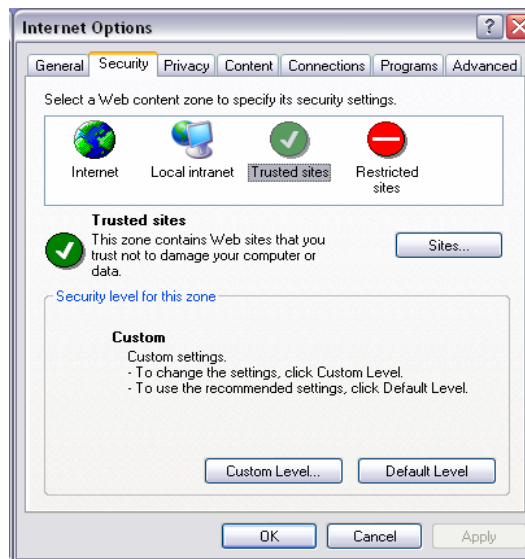
Steps

To configure internet Internet Explorer to recognize the Hammer XMS server as a *trusted site*:

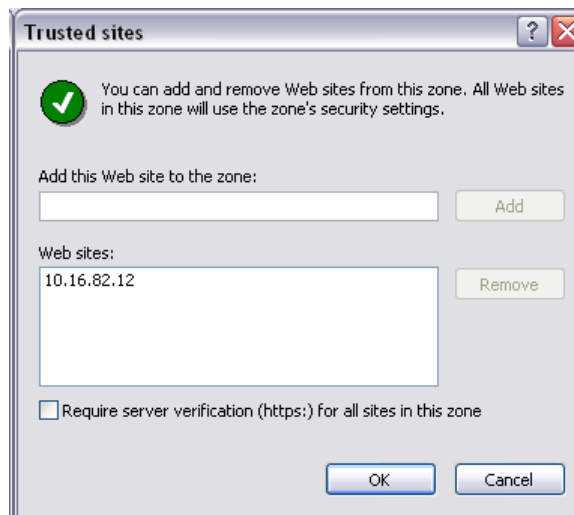
1. Open Internet Explorer and click on Tools-> Internet Options. The following dialog will display.



2. Click the *Security* tab, click the *Trusted Sites* icon, and click on the *Sites* button.



The following dialog will appear. Make sure the Check box *Require server verification(https:) for all sites in this zone* is unchecked before continuing.

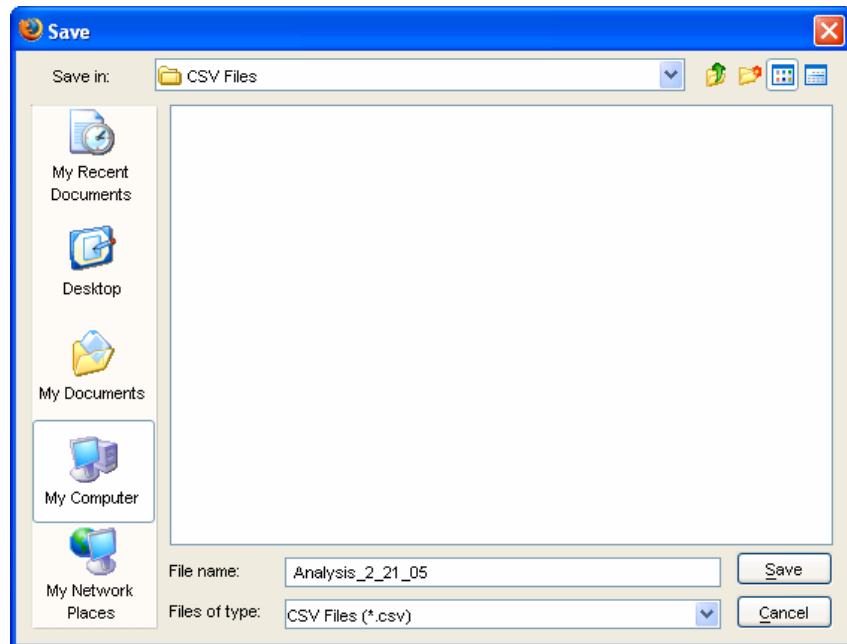


3. In the *Add this website to the zone text* field, enter the web address of the XMS web server.
4. Click the add button (enabled once text is entered).

Steps

To Generate and View Hammer XMS call data in CSV file format:

1. At the main Hammer XMS page select *Analysis*.
2. Select the report type, tag (if applicable), and time interval.
3. Right-click the cell containing the data you wish to generate, and select *Save Table to a CSV File*. A save dialog will display, as shown in the following figure.



4. Save the data to the desired location on your remote client.

5. Locate, and double-click the file to view the file data in spreadsheet format, as shown in the following figure.

Microsoft Excel - Analysis_2_21_05.csv

File Edit View Insert Format Tools Data RoboPDF Window Help Adobe PDF

Type a question for help

SnagIt Window

A1	Hour	B	C	D	E	F	G	H	I	J	K	L	M	N	O	
1	Hour	Atmpt	Call	Seize Rati	Comp Call	Failed	% Failed	Calls/Sec	Dur-Avg	Dur-Min	Dur-Max	PPD-Avg	PPD-Min	PPD-Max	Con-Avg	Con-Min
2	Mon Feb 2	619	97	620	19	3	0	12	3	60	10	7	97	13	2	
3	Mon Feb 2	605	97	604	18	3	0	10	3	63	10	7	102	13	5	
4	Mon Feb 2	682	98	681	16	2	0	15	3	120	9	7	76	12	5	
5	Mon Feb 2	615	98	617	10	2	0	11	3	120	8	7	74	11	6	
6	Mon Feb 2	583	98	581	10	2	0	11	3	63	9	7	101	11	7	
7	Mon Feb 2	711	99	710	9	1	0	9	3	62	10	7	91	11	5	
8	Mon Feb 2	557	98	560	10	2	0	13	3	120	9	7	68	11	8	
9	Mon Feb 2	986	98	985	18	2	0	15	3	120	9	7	87	12	5	
10	Mon Feb 2	1020	97	1018	34	3	0	16	3	120	9	7	112	13	5	
11	Mon Feb 2	967	97	968	27	3	0	16	3	120	9	7	118	13	5	
12	Mon Feb 2	1007	97	1007	29	3	0	16	3	120	9	6	101	13	8	
13	Mon Feb 2	958	97	958	27	3	0	17	3	120	9	7	96	13	8	
14	Mon Feb 2	61	93	61	4	7	0	21	3	120	7	7	12	12	8	

Ready

Analysis_2_21_05/

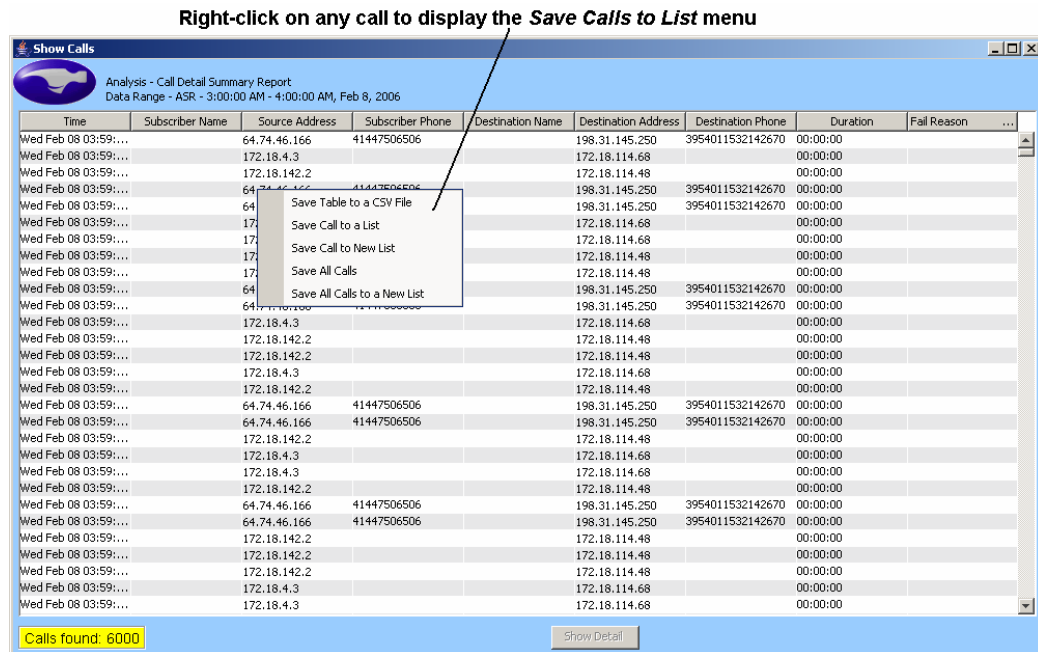
Using the Show Calls Feature

Access the *Show Calls* function by right-clicking the mouse in the right panel table cell of any of the active reports for all column data except *CPS*, *ASR*, *% Calls Failed* or *Avg. Values*, and selecting *Show Calls*. A new window, similar to the one in the following figure, will display. For *Min* and *Max* columns the *Show Calls* window will return call data with the Max value (or near the Max Value) or call data with the Min value (or near the Min Value) respectively.

Example: Using the *Show Calls* feature within a *Call Summary Active Report*:

- The user sees a suspicious value for *Max Call Length* on the 4th of Oct. 05:00 EDT row of the report.
- The user right clicks on the cell and selects the *Show Calls* menu item.
- A *Show Calls* window appears.

- Because the *Active Report* and *Show Calls* windows are non-modal, the user can move freely between the two.
- The user is free to create as many Show Calls windows as required to investigate the problem.
- The user can save any or all of the calls displayed to a call list by right-clicking on any of the calls.

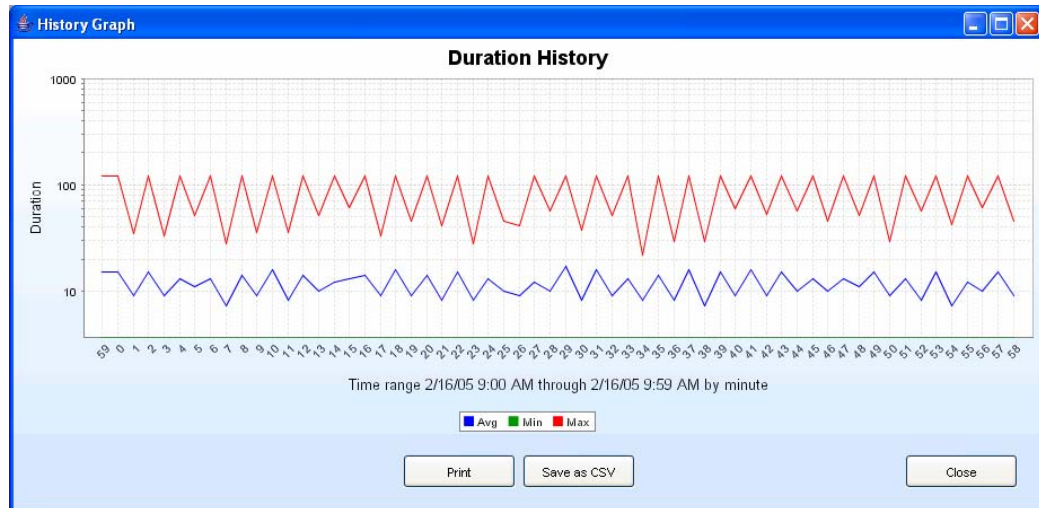


Understanding *History Graphs*

History Graph data, shown in the following figure, has different purposes depending on the *Active Report* and *Column* selected, when invoking the graph. In general, use the graph to view specific tracking data over a selected time period. The History Graphs feature allows the user to:

- View tracking data in the form of Min, Max and Average where appropriate. The graph displays three lines when all three data items are available.

- Generate historical call data (from the right click menu) from most of the Active Report columns.
- Move back and forth between report and graph windows, and create more than one graph.



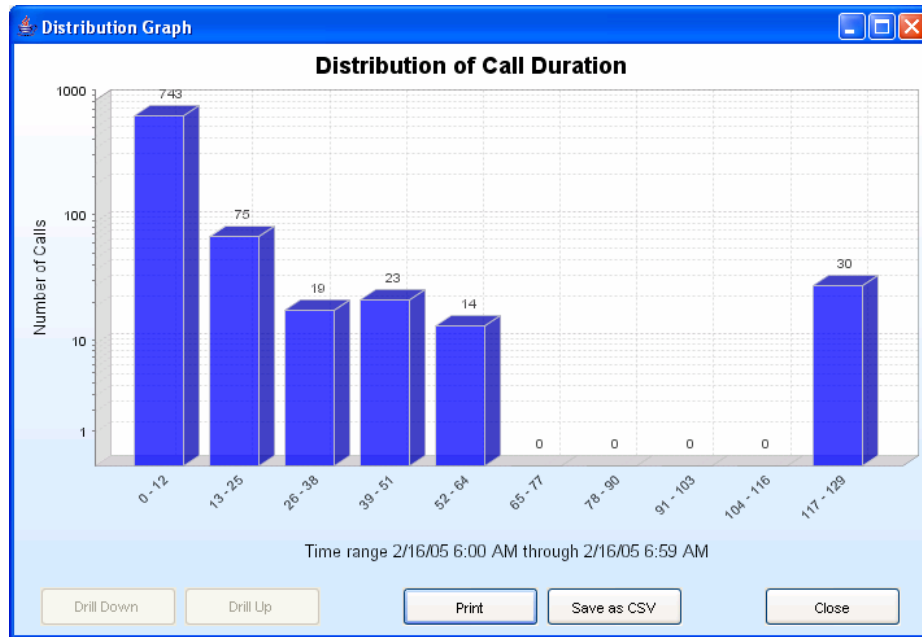
Understanding *Distribution Graphs*

Use Distribution graphs, a sample of which is shown in the following figure, to quickly identify and analyze data that contains a wide spectrum of values.

The Distribution Graph feature includes the following features:

- The ability to see the distribution of calls that make up the data for a given metric over a user-defined number of buckets (bars on the graph).
- Distribution Graphs use a logarithmic vertical axis because a typical dataset contains a wide range of values, varying by orders of magnitude from smallest to largest.
- Distribution Graphs can be invoked (using the right click menu) from a subset of columns within the Summary related Active Reports.

- The user can freely move back and forth between screens and create more than one graph.
- Select any of the graph bars, and use the Drill Down button to break down further the range of values represented in that bar. Use the Drill Up button to return to the previous, wider distribution.



Creating Reports

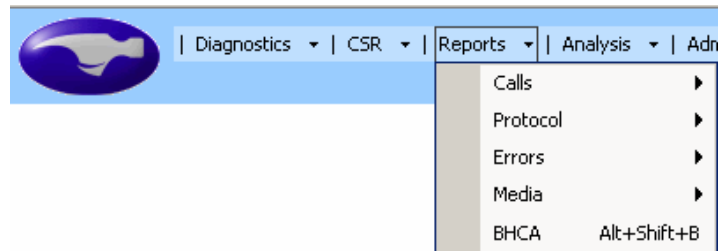
The Hammer XMS Reports suite provides a set of predefined reports that provide those VoIP metrics most commonly needed by carrier engineering staff or management charged with testing, repairing, and maintaining VoIP networks. The reports included in Hammer XMS version 1.5 include:

- Call Summary
- Call Detail Summary
- Errors By Type Summary
- VoIP Media Statistics Summary
- VoIP MGCP Media Statistics Summary
- SIP Protocol Summary
- SS7 Protocol Summary
- MGCP Protocol Summary
- MEGACO Protocol Summary
- H.323 Protocol Summary
- ISDN Protocol Summary
- BHCA (Busy-Hour Call Analysis)

Steps

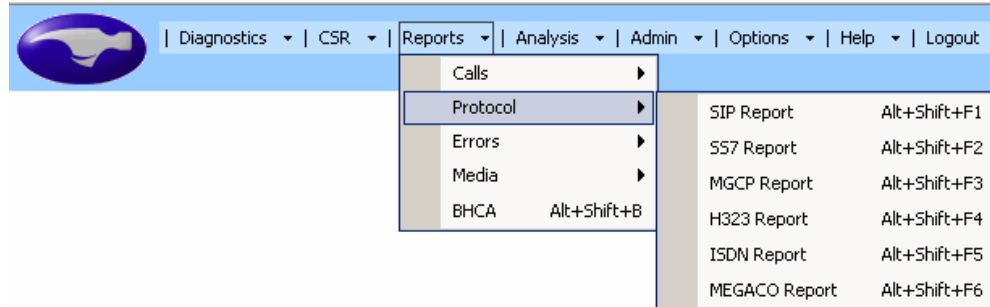
To create a report:

1. At the main Hammer XMS Web Page, click the *Reports* link. The Reports drop-down menu will display, as shown in the following figure.

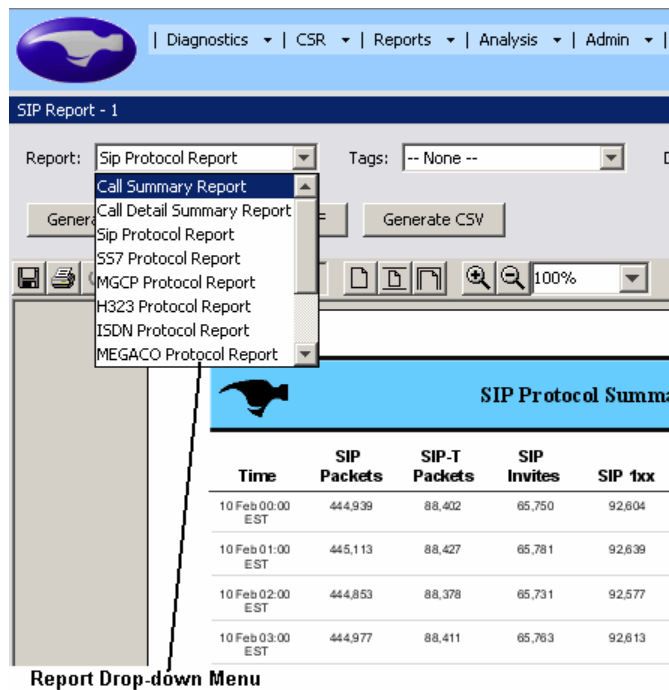


2. Mouse-over any of the first four report types to select a specific report within that report type.

Example: Mousing over the *Protocol* report type displays all of the protocol reports available, as shown following.



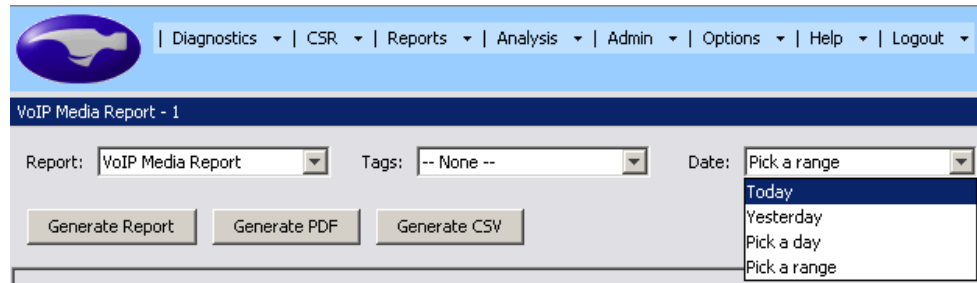
3. Once you have selected and displayed a report (see the following figure), use the Report drop-down menu to select any additional reports.



Selecting Time Intervals for Reports

Each report features a Time interval drop-down menu, shown in the following figure, with the following interval options:

- Today
- Yesterday
- Pick a Day
- Pick a Range



When a single day is selected (Today / Yesterday), the report metrics will be broken down by hours. When a week interval is (This Week / Last Week) selected, the report metrics will be broken down by day.

Pick A Range

The Pick A Range feature gives you the ability to retrieve data using the start and end dates of your choice.

Steps

To select start and end dates for your data search:

1. Click the Pick A Range link. The Pick A Range dialog will appear, as shown in the following figure.

Date Range

Pick The Start Time:

< January > << 2005 >>

Mon	Tue	Wed	Thu	Fri	Sat	Sun
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

Pick The End Time:

< February > << 2005 >>

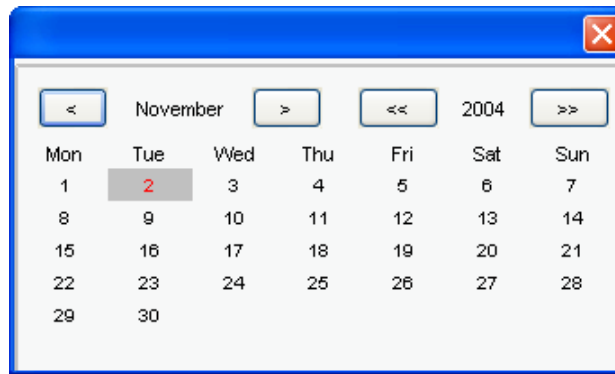
Mon	Tue	Wed	Thu	Fri	Sat	Sun
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28						

OK Cancel

2. Use the forward/backward arrows to select the month values. Use the forward/backward double arrows to select year values.
3. Click to select a start date from the upper half of the dialog, and click an end date from the lower half.
4. Click OK when start and end dates have been selected.

Pick a Day

When clicked, the Pick A Day option displays the following dialog. The current day is displayed in red.



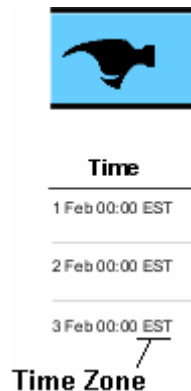
To use the Pick a Day dialog:

1. At the top of the dialog, select the month and year, using the increment/decrement arrow buttons for each.
2. Click the desired day.

Understanding Report Time Zones

The time zone displayed in any report you generate will reflect the time zone to which your client computer is set at the time the report is generated (shown below).

Thus, if the client on which you've requested the report is set to Eastern Standard Time (EST), your report will show all call times as EST.



The possible time zones and their abbreviations are as follows:

Region	Daylight Savings	Standard Time
Eastern	EDT	EST
Central	CDT	CST
Mountain	MDT	MST
Pacific	PDT	PST

Important! When creating Busy Hour Call Analysis (BHCA) reports, it's important to remember that the *repprep* process retrieves the data from two different sources, depending on the interval for which the query is made.

- When requesting Busy Hour Call data for intervals prior to today, the data is derived from the database *Call Table*. This process captures data for only those calls that are *completed*.
- When requesting Busy Hour Call data for today, the data is derived from the database *Tracking Call Process*. This process captures data for all calls *initiated*.

Call Summary Report

The Call Summary Report, shown in the figure, below, provides call volume statistics.

Call Summary Report for Monday 13 February 2006						
Time	Calls Attempted	ASR	NER	Calls Completed	% Calls Failed	Avg. Call Length
13 Feb 00:00 EST	274,524	100	100	274,515	<1%	00:04:45
13 Feb 01:00 EST	218,755	100	33	218,764	<1%	00:02:20
13 Feb 02:00 EST	216,696	100	27	216,683	<1%	00:06:44
13 Feb 03:00 EST	272,014	100	30	271,997	<1%	00:20:05
13 Feb 04:00 EST	339,011	100	27	339,014	<1%	00:19:19
13 Feb 05:00 EST	331,619	100	31	331,607	<1%	00:14:08
13 Feb 06:00 EST	273,267	100	33	273,281	<1%	00:56:16
13 Feb 07:00 EST	307,434	100	26	307,434	<1%	00:33:07
13 Feb 08:00 EST	270,861	100	30	270,856	<1%	00:09:42
13 Feb 09:00 EST	314,797	100	34	314,780	<1%	00:48:00
13 Feb 10:00 EST	338,762	100	20	338,748	<1%	00:19:25
13 Feb 11:00 EST	336,918	100	10	336,924	<1%	00:27:07
13 Feb 12:00 EST	338,930	100	10	338,928	<1%	00:51:00
13 Feb 13:00 EST	303,009	100	10	303,010	<1%	00:00:01
13 Feb 14:00 EST	266,486	100	10	266,481	<1%	00:00:01
13 Feb 15:00 EST	305,533	100	30	305,462	<1%	00:08:00
13 Feb 14:00 EST	266,486	100	26	266,481	<1%	00:00:01
13 Feb 15:00 EST	305,533	100	27	305,462	<1%	00:08:00
13 Feb 16:00 EST	340,805	100	21	340,826	<1%	00:10:05
13 Feb 17:00 EST	205,268	100	21	205,254	<1%	00:50:34
Summary Min, Max and Totals:						
	5,254,689	100	100	5,254,564	<1%	00:21:25

A list and description of the metrics included in the report is provided in the following table.

Metric	Description
Time	Time interval selected (Yesterday, Today, Pick a Range, Pick A Day.
Calls Attempted	Total number of calls attempted during the interval

Metric	Description
NER (Network Efficiency Ration)	Answers + User Busy + Ring No Answer + Terminal Rejects, divided by the total call attempts (seizures).
ASR	The Answer Seize Ratio expressed as a percentage of calls that were successfully connected, compared to calls attempted, during the interval.
Calls Completed	Total number of calls completed during the interval.
% Calls Failed	The percentage of calls that failed, of all the calls completed, during the interval.
Avg. Call Length	The average call length for calls that completed during the interval.

Call Detail Summary Report

The Call Detail Summary Report provides a more detailed look at call traffic statistics (see the following figure).

Call Detail Summary Report for Monday 13 February 2006																	
Time	Calls Attempted	ASR	NER	Calls Completed	Calls Failed	% Calls Failed	Avg CPS	Avg Call Length	Min Call Length	Max Call Length	Avg PDD (ms)	Min PDD (ms)	Max PDD (ms)	Avg CL (ms)	Min CL (ms)	Max CL (ms)	Avg DT (ms)
13 Feb 00:00 EST	274,524	100	21	274,515	535	<1%	76	00:04:45	00:00:01	49:17:02:47	238	1	7,095	182	1	7,095	262
13 Feb 01:00 EST	218,755	100	27	218,764	456	<1%	61	00:02:20	00:00:01	49:17:02:47	317	1	9,446	211	1	11,835	413
13 Feb 02:00 EST	216,896	100	27	216,883	410	<1%	60	00:06:44	00:00:01	49:17:02:47	342	1	8,884	209	1	11,697	406
13 Feb 03:00 EST	272,014	100	30	271,997	520	<1%	75	00:20:05	00:00:01	49:17:02:47	234	1	7,367	184	1	10,207	278
13 Feb 04:00 EST	339,011	100	34	339,014	579	<1%	94	00:19:19	00:00:01	49:17:02:47	132	1	2,555	166	1	2,289	112
13 Feb 05:00 EST	331,619	100	15	331,607	577	<1%	789	00:14:08	00:00:01	49:17:02:47	135	1	5,363	167	1	9,023	127
13 Feb 06:00 EST	273,267	100	27	273,281	555	<1%	75	00:56:16	00:00:01	49:17:02:47	231	1	10,539	187	1	13,444	319
13 Feb 07:00 EST	307,434	100	31	307,434	555	<1%	85	00:33:07	00:00:01	49:17:02:47	181	1	4,289	170	1	4,337	184
13 Feb 08:00 EST	270,861	100	33	270,856	538	<1%	75	00:09:42	00:00:01	49:17:02:47	229	1	5,401	183	1	8,514	267
13 Feb 09:00 EST	314,797	100	33	314,780	557	<1%	87	00:48:00	00:00:01	49:17:02:47	162	1	5,437	170	1	7,476	163
13 Feb 10:00 EST	338,762	100	33	338,748	583	<1%	94	00:19:25	00:00:01	49:17:02:47	132	1	2,549	166	1	2,301	113
13 Feb 11:00 EST	336,918	100	30	336,924	572	<1%	93	00:27:07	00:00:01	49:17:02:47	131	1	3,500	165	1	2,663	118
13 Feb 12:00 EST	338,930	100	34	338,928	580	<1%	94	00:51:00	00:00:01	49:17:02:47	132	1	2,549	164	1	2,287	116
13 Feb 13:00 EST	303,009	100	15	303,010	563	<1%	84	00:00:01	00:00:01	00:01:05	190	1	5,345	174	1	6,062	199
13 Feb 14:00 EST	266,486	100	13	266,481	521	<1%	74	00:00:01	00:00:01	00:01:34	237	1	7,501	187	1	8,558	276
13 Feb 15:00 EST	305,533	100	13	305,462	664	<1%	86	00:08:00	00:00:01	49:17:02:47	158	1	6,939	172	1	11,103	170
13 Feb 16:00 EST	340,805	100	26	340,826	583	<1%	94	00:10:05	00:00:01	49:17:02:47	132	1	2,549	166	1	2,289	114
13 Feb 17:00 EST	153,991	100	30	153,979	263	<1%	95	00:22:31	00:00:01	49:17:02:47	132	1	2,549	166	1	2,287	113
Summary, Min, Max and Totals:																	
	5,203,412	null	null	5,203,289	9,611	<1%	87	00:02:01	XXX	00:00:01	185	1	10,539	176	1	13,444	198

Refer to the following table for a list and description of the call detail metrics contained in this report.

Metric	Description
Time	Time interval selected (Yesterday, Today, Pick a Range, Pick A Day.
Calls Attempted	Total number of calls attempted for the selected interval
ASR (Answer Seize Ratio)	The Answer Seize Ratio expressed as a percentage of calls that were successfully connected, compared to calls attempted, during the interval.
NER (Network Efficiency Ratio)	Answers + User Busy + Ring No Answer + Terminal Rejects, divided by the total call attempts (seizures).
Calls Completed	Total number of calls completed during the interval.
Calls Failed	Total number of calls that failed during the interval
% Calls Failed	The percentage of calls that failed, of all the calls completed, during the interval.
Average CPS	Average Calls Per Second over the interval, i.e. calls attempted / interval period (in seconds).
Average Call Length	The average call length for calls that completed during the interval, in seconds.
Min Call Length	The minimum call length for calls that completed during the interval.
Max call length	The maximum call length for calls that completed during the interval.
Average PDD	The average Post Dial Delay, in ms.
Min PDD	The minimum post dial delay.
Max PDD	The maximum post dial delay.
Average CL	The average Connect Latency in ms.
Min CL	The minimum Connect Latency.
Max CL	The maximum Connect Latency.
Average DT	The average Disconnect Time in ms.
Min DT	The minimum Disconnect Time.

Metric	Description
Max DT	The maximum Disconnect Time.

Errors by Type Summary Report

The Errors by Type Summary, shown in the following figure, provides call error totals by error type for the protocols your system is configured to support.

As shown, the daily totals for each MGCP error codes are displayed. In this example, the search period was the month of July, thus the summaries for each code are broken down by day.

Set 9 of 11													
Time	MGCP 511	MGCP 512	MGCP 513	MGCP 514	MGCP 515	MGCP 516	MGCP 517	MGCP 518	MGCP 519	MGCP 520	MGCP 521	MGCP 522	MGCP 523
1 Jul 00:00 EDT	3	14	1	10	14	9	8	6	11	5	11		2
2 Jul 00:00 EDT	9		22	8	1	8	14	15	9	9	14		8
3 Jul 00:00 EDT		3	9	8	8	1		3	1	4	5	5	15
4 Jul 00:00 EDT	9	7	7	4	18	6	7	5	10	4		13	1
5 Jul 00:00 EDT	5		5	10	6	9		10		8	6	9	4
6 Jul 00:00 EDT	11	3	9	11	6	4	12	1	10	5	7	19	6
7 Jul 00:00 EDT	2	7	1	3	9		3		6		5	3	2
8 Jul 00:00 EDT	10		2	11	11	9	10	8			5	23	10
9 Jul 00:00 EDT	6	9	6	12	4	5	5	5	7	4	6	11	12
10 Jul 00:00 EDT	3		7	6	10	11	9	12	5	12	5	13	
11 Jul 00:00 EDT	9	8	13	3	12	4	9	5	1	6	7		6
12 Jul 00:00 EDT		10	5	4	8	2	6	9		9	10		
13 Jul 00:00 EDT	1	2	6	4		3	1		2	5	5	17	7
14 Jul 00:00 EDT	10	18	2		2	3	18	6	2	10	7	15	14
15 Jul 00:00 EDT	9	6	4	9	7		17	3	4	4	5	1	4
16 Jul 00:00 EDT	9		20	10		2	4	13	7	2			5
17 Jul 00:00 EDT	4	3	3	6	2	10	7	10	3	1	17	10	20
18 Jul 00:00 EDT			13	2	8	11	1	4	7		9	15	
19 Jul 00:00 EDT	14	8	7	3	1	5		5	6	4	7	4	10
20 Jul 00:00 EDT	5	3	11	10	7	12	9	7	11	3	9	4	8
21 Jul 00:00 EDT	5	6	12	2	12	2	4			6	5	8	
22 Jul 00:00 EDT	1	4			4	4			5	5	5		3
23 Jul 00:00 EDT	4	12	2	11	22	13	6				7	5	4
24 Jul 00:00 EDT	4	1	7	2	8	4	13	8		6			6
25 Jul 00:00 EDT		2	7			5	4			2	13	1	2
26 Jul 00:00 EDT	8		4		6	1	7	11	5	20	10	16	4
27 Jul 00:00 EDT	12	7	1	2	5	2	1	5	14	6	3	4	11
28 Jul 00:00 EDT		6	14	7		11	3		3	17	8	4	9
29 Jul 00:00 EDT		4	3				2	14	1	3	5	4	5

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Error summaries are displayed, one protocol at a time, across multiple pages, as shown in the following figure.

Adobe Reader - [ErrorByTypeReport58622[1].pdf]

File Edit View Document Tools Window Help

Save a Copy Search Select 73% Help

Options x

Pages

1

2

3

4

Attachments

Comments

Errors by Type Summary Report for Friday 01 July 2005 - Monday 01 August 2005

Set 1 of 11

Time	SIP 400	SIP 401	SIP 402	SIP 403	SIP 404	SIP 405	SIP 406	SIP 407	SIP 408	SIP 409	SIP 410	SIP 411	SIP 413
1 Jul 00:00 EDT	6	4	9	1	8	1	10	7	2	7	8	13	4
2 Jul 00:00 EDT	4	20		5	4		5	7	2	3	5	1	5
3 Jul 00:00 EDT	2		9	8	18	2	8	5	7	7	13	3	13
4 Jul 00:00 EDT	2	6	1	7	11	14	9		11	8	8	6	5
5 Jul 00:00 EDT		4	3	8	6	4	10	3	3	9	4	6	
6 Jul 00:00 EDT	4	4	5		8	5	1	1		5	19		2
7 Jul 00:00 EDT	17	10	5			4	3	4	3	12	5	1	
8 Jul 00:00 EDT	2	9	12	5	13	4	5	16		5	10	5	5
9 Jul 00:00 EDT	7	14			2		4		1	15		2	
10 Jul 00:00 EDT	5	2	3		17	14	2		3	1		8	5
11 Jul 00:00 EDT	12	2		2	4		7	16	2	5		5	8
12 Jul 00:00 EDT		1	14	3		8	5	8	6	4	3	6	5
13 Jul 00:00 EDT		10	5	4	4	8		3	9	6		1	5
14 Jul 00:00 EDT		7	3		1	5		6	3	5		4	6
15 Jul 00:00 EDT	3	2	10	11		3	8	8	12	4		1	
16 Jul 00:00 EDT	4		1	2		22	6	5	3	8	7	3	5
17 Jul 00:00 EDT	12	1	6		2	8	4	2	15	5	3		2
18 Jul 00:00 EDT	4	6		7	1	4	12	2	5	13			8
19 Jul 00:00 EDT	9		20	5	13	9		4	3		5	5	17
20 Jul 00:00 EDT	12	3	5	16	5	9	4	1	3	8	8	12	
21 Jul 00:00 EDT	5	6	9		4	14	4		16		9		4
22 Jul 00:00 EDT	1	3		9	9	7	6		5	4	3	14	15
23 Jul 00:00 EDT	7	3		18	11	4	12	7	4	5	1	10	3
24 Jul 00:00 EDT	18	11	2	4	14	3		2	4	1		7	3
25 Jul 00:00 EDT	6	9		15		3	8	4	5	8	20	3	1
26 Jul 00:00 EDT	11	12	4		10	15	2	12		7			7

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VoIP Media Statistics Report

The VoIP Media Statistics Report, shown below, provides useful information on media packet metrics, including Packets and Octets received, packets lost, Jitter, and Latency, R-factor and MOS score.

Understanding Voice Quality Statistics

It's important to remember that there is some correlation between voice quality, and jitter and R-factor metrics – measurements that are reported in the VoIP Media Statistics Report. When jitter is present in an RTP stream, the voice quality goes down when someone listens to the raw RTP stream. In general, if an RTP stream exhibits very high jitter, it will result in poor R-Factor and poor voice quality.

In order to reduce the effect of jitter in an RTP stream, VoIP network systems typically send the stream through a jitter buffer, which causes a smoothing of the stream. This results in improved voice quality, though buffering the stream also adds some latency to the individual packets.

The key is maintaining enough jitter buffer to effectively smooth out an RTP stream exhibiting high jitter. As long as the jitter buffer is bigger than the jitter in the stream, the R-Factor and voice quality of the resulting RTP stream remains good.

VoIP network equipment seeks to balance buffering and latency factors to determine an optimal size for jitter buffer that insures the best possible voice quality, with the lowest possible end-to-end latency.



VoIP Media Statistics Report for Thursday 08 September 2005

Time	Packets Received	Octets Received	Dup Packets	Packets Lost	% Packet Lost	Packets OOS	Avg Jitter	Min Jitter	Max Jitter	Avg R Factor	Min R Factor	Max R Factor	Avg MOS	Min MOS	Max MOS
8 Sep 00:00 EDT	492,659	107,399,662	0	40	<1%	0	3	3	4	92	92	92	4.1797	4.1797	4.1797
8 Sep 01:00 EDT	502,896	109,631,328	0	43	<1%	0	3	3	4	92	92	92	4.1797	4.1797	4.1797
8 Sep 02:00 EDT	505,570	110,214,260	0	52	<1%	0	3	3	4	92	92	92	4.1797	4.1797	4.1797
8 Sep 03:00 EDT	364,180	79,391,240	0	50	<1%	0	3	3	4	92	92	92	4.1797	4.1797	4.1797
8 Sep 04:00 EDT	363,654	79,276,572	0	53	<1%	0	3	3	4	92	92	92	4.1797	4.1797	4.1797
8 Sep 05:00 EDT	358,959	78,253,062	0	50	<1%	0	3	3	4	92	92	92	4.1797	4.1797	4.1797
8 Sep 06:00 EDT	444,323	96,862,414	0	73	<1%	0	3	3	4	92	92	92	4.1797	4.1797	4.1797
8 Sep 07:00 EDT	953,691	207,904,638	0	86	<1%	0	3	3	4	92	92	92	4.1797	4.1797	4.1797
8 Sep 08:00 EDT	996,463	217,228,934	0	117	<1%	0	3	3	4	92	92	92	4.1797	4.1797	4.1797
8 Sep 09:00 EDT	995,007	216,911,526	0	118	<1%	0	3	3	4	92	92	92	4.1797	4.1797	4.1797
8 Sep 10:00 EDT	987,042	215,175,196	0	107	<1%	0	3	3	4	92	92	92	4.1797	4.1797	4.1797
8 Sep 11:00 EDT	995,167	216,946,406	0	116	<1%	0	3	3	4	92	92	92	4.1797	4.1797	4.1797
8 Sep 12:00 EDT	1,187,340	252,300,120	0	152	<1%	0	3	3	4	92	92	92	4.1797	4.1797	4.1797
8 Sep 13:00 EDT	1,162,539	253,433,502	0	141	<1%	0	3	3	4	92	92	92	4.1797	4.1797	4.1797
8 Sep 14:00 EDT	1,169,461	254,942,498	0	156	<1%	0	3	3	4	92	92	92	4.1797	4.1797	4.1797
8 Sep 15:00 EDT	1,173,702	255,867,036	0	157	<1%	0	3	3	4	92	92	92	4.1797	4.1797	4.1797
8 Sep 16:00 EDT	1,011,386	220,482,148	0	123	<1%	0	3	3	4	92	92	92	4.1797	4.1797	4.1797
8 Sep 17:00 EDT	986,910	215,146,380	0	107	<1%	0	3	3	4	92	92	92	4.1797	4.1797	4.1797

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The following table lists and describes the metrics included in this report.

Metrics	Description
Time	Time interval selected (Yesterday, Today, Pick a Range, Pick A Day.
Packets Received	The total number of packets received.
Octets Received	The total number of bytes received.
Dup Packets	The total number of duplicate packets received.
Packets Lost	The total number of packets lost
% of Packets Lost	The percentage of packets lost

Packets OOS	The total number of 'Out Of Sequence' packets received for the for the interval.
Avg Jitter	The average jitter reported for all samples ² taken during each time unit of the time interval (each hour of a day, or each day of the week).
Min Jitter	The minimum jitter reported for the interval.
Max Jitter	The maximum jitter reported for the interval.
Avg R Factor	The average mathematically-derived voice quality value for the interval, between 0 and 100 where: >90 = All users very satisfied 80 – 89 = All users satisfied 70 – 79 = Some users dissatisfied 60 – 69 = Many users dissatisfied <60 = Nearly all users dissatisfied
Min R Factor	The minimum voice quality rating for the reported interval.
Max R Factor	The maximum voice quality rating for the reported interval.
Avg MOS	The average Mean Opinion Score (user-perceived voice quality based on subjective testing) for the reported interval.
Min MOS	The minimum Mean Opinion Score (user-perceived voice quality based on subjective testing) for the reported interval.
Max MOS	The maximum Mean Opinion Score (user-perceived voice quality based on subjective testing) for the reported interval.

² A 'sample' is defined as a completed call that includes a 250 response to the DLCX request.

VoIP MGCP Media Statistics Report

The VoIP MGCP Media Statistics Report provides a useful view of MGCP packet activity, including packet and octet totals, plus jitter and latency statistics.

VoIP MGCP Media Statistics Report for Sunday 04 September 2005 - Friday 09 September 2005										
Time	Packets Received	Octets Received	Packets Lost	% Packets Lost	Jitter	Min Jitter	Max Jitter	Latency	Min Latency	Max Latency
7 Sep 00:00 EDT	348,651,719	27,891,518,092	2,043,665	1	1,026	0	6,725,899	6	0	120
8 Sep 00:00 EDT	1,454,529	116,362,120	195,081	12	0	0	2	4	0	10
Summary Min, Max and Totals:										
	350,106,248	28,007,880,212	2,238,746	1	1,026	0	6,725,899	6	0	120

The following table lists and describes the metrics included in this report.

Metrics	Description
Time	Time interval selected (Yesterday, Today, Pick a Range, Pick A Day.
Packets Received	The total number of packets received during the selected interval.
Octets Received	The total number of bytes received during the selected interval.
Packets Lost	The total number of packets lost during the selected interval.
% of Packets Lost	The percentage of packets lost during the selected interval.
Jitter	The average jitter reported for all samples ³ taken during each time unit of the time interval (each hour of a day, or each day of the week).
Min Jitter	The minimum jitter reported for the

³ A 'sample' is defined as a completed call that includes a 250 response to the DLCX request.

	selected interval.
Max Jitter	The maximum jitter reported for the interval.
Latency	The delay, in milliseconds, of actual packet arrival from expected packet arrival.
Min Latency	The minimum delay, in milliseconds, of actual packet arrival from expected packet arrival.
Max Latency	The maximum delay, in milliseconds, of actual packet arrival from expected packet arrival.

SIP Protocol Summary Report

The SIP Protocol Summary Report, shown in the following figure, provides metrics on SIP signaling received at the Hammer XMS, including response class messaging.

SIP Protocol Summary Report for Thursday 09 September 2004											
Time	SIP Packets	SIP-T Packets	SIP Invites	SIP 1xx	SIP 2xx	SIP 3xx	SIP 4xx	SIP 5xx	SIP Bye	SIP Register	SIP Ack
9 Sep 00:00 EDT	740	0	3	7	355	0	0	0	3	0	3
9 Sep 01:00 EDT	713	0	0	0	349	0	0	0	0	0	0
9 Sep 02:00 EDT	845	10	24	34	376	0	0	12	10	0	23
9 Sep 03:00 EDT	841	15	27	33	367	0	0	19	10	0	27
9 Sep 04:00 EDT	722	0	0	0	354	0	0	0	0	0	0
9 Sep 05:00 EDT	720	0	0	0	353	0	0	0	0	0	0
9 Sep 06:00 EDT	733	0	0	0	359	0	0	0	0	0	0
9 Sep 07:00 EDT	726	0	1	1	356	0	0	0	1	0	1
9 Sep 08:00 EDT	760	0	16	3	364	0	1	0	1	0	2
9 Sep 09:00 EDT	806	0	18	13	377	0	6	0	5	0	11
9 Sep 10:00 EDT	942	4	47	45	412	0	7	1	21	0	33
9 Sep 11:00 EDT	772	4	12	5	369	0	0	0	5	0	5

The following table lists and describes the metrics included in this report.

Metric	Description
Time	Time interval selected (yesterday, today, this week, last week).
SIP Packets	Total number of SIP packets received during the selected interval.
SIP-T Packets	Total number of SIP-T packets received during the selected interval.
SIP Invites	Total number of SIP Invites received during

Metric	Description
	the selected interval.
SIP 1xx	Total number of SIP response 1xx (informational) messages received during the selected interval.
SIP 2xx	Total number of SIP response 2xx (successful response) messages received during the selected interval.
SIP 3xx	Total number of SIP response 3xx (redirection response) messages received during the selected interval.
SIP 4xx	Total number of SIP response 4xx (request failure response) messages received during the selected interval.
SIP 5xx	Total number of SIP response 5xx (server failure response) messages received during the selected interval.
SIP Register	Total number of SIP Register messages received during the selected interval.
SIP Ack	Total number of SIP Acks received during the selected interval.

SS7 Protocol Summary Report

The SS7 Protocol Summary Report, shown in the following figure, provides data on SS7 packet streams.

SS7 Protocol Summary Report for Monday 01 August 2005 - Tuesday 13 September 2005												
Time	SS7 Packets	SS7 LSSU	SS7 MSU	SS7 SNM	SS7 MTNS	SS7 IAM	SS7 ACM	SS7 ANM	SS7 REL	SS7 RLC	TCAP Requests	TCAP Responses
12 Sep 00:00 EDT	72144	0	72144	21	561	14809	12237	10650	13852	13730	0	0
Summary Min, Max and Totals:												
	72144	0	72144	21	561	14809	12237	10650	13852	13730	0	0

The following table lists and describes the metrics included in this report.

Metric	Description
Time	Time interval selected (yesterday, today, this week, last week).
SS7 Packets	Total number of SS7 packets received during the selected interval.
SS7 LSSU	Total number of SS7 Link Status Signal Unit messages received during the selected interval.
SS7 MSU	Total number of SS7 Message Signal Unit messages received during the selected interval.
SS7 SNM	Total number of SS7 Signaling Network Management messages received during the selected interval.
SS7 MTNS	Total number of SS7 Maintenance Special messages received during the selected interval.

Metric	Description
SS7 IAM	Total number of SS7 Initial Address messages received during the selected interval.
SS7 ACM	Total number of SS7 Address Complete messages received during the selected interval.
SS7 ANM	Total number of SS7 Answer messages received during the selected interval.
SS7 REL	Total number of SS7 Release messages received during the selected interval.
SS7 RLC	Total number of SS7 RLC messages received during the selected interval.
TCAP Requests	Total number of TCAP Request messages received during the selected interval.
TCAP Responses	Total number of TCAP Response messages received during the selected interval.

MGCP Protocol Summary Report

The MGCP Protocol Summary Report, shown in the following figure, provides data on media packet streams.

MGCP Protocol Summary Report for Tuesday 24 August 2004								
Time	MGCP Packets	MGCP Notify	MGCP CRCX	MGCP MDCX	MGCP DLCX	MGCP RQNT	MGCP AUER	MGCP Errors
24 Aug 00:00	0	0	0	0	0	0	0	0
24 Aug 01:00	0	0	0	0	0	0	0	0
24 Aug 02:00	0	0	0	0	0	0	0	0
24 Aug 03:00	0	0	0	0	0	0	0	0
24 Aug 04:00	0	0	0	0	0	0	0	0
24 Aug 05:00	0	0	0	0	0	0	0	0
24 Aug 06:00	0	0	0	0	0	0	0	0
24 Aug 07:00	0	0	0	0	0	0	0	0
24 Aug 08:00	0	0	0	0	0	0	0	0
24 Aug 09:00	0	0	0	0	0	0	0	0
24 Aug 10:00	0	0	0	0	0	0	0	0
24 Aug 11:00	0	0	0	0	0	0	0	0
Summary Min, Max and Totals:								

The following table lists and describes the metrics included in this report.

Metric	Description
Time	Time interval selected (yesterday, today, pick a day, pick a range).
MGCP Packets	Total number of MGCP packets over the interval.
MGCP Notify	Total number of MGCP Notify messages over the interval.

Metric	Description
MGCP CRCX	Total number of MGCP Create Connection messages over the interval.
MGCP MDCX	Total number of MGCP Modify Connection messages over the interval.
MGCP DLCX	Total number of MGCP Delete Connection messages over the interval.
MGCP RQNT	Total number of MGCP Request Notification messages over the interval.
MGCP AUPE	Total number of MGCP Audit Endpoint messages over the interval.
MGCP Errors	Total number of MGCP error messages over the interval.

MEGACO Protocol Summary

The MEGACO Protocol Summary Report, shown in the following figure, provides data on media packet streams.

MEGACO Protocol Summary Report for Tuesday 14 February 2006					
Time	MEGACO Packets	ME GACO Adds	ME GACO Subtracts	MEGACO Modifys	ME GACO Errors
14 Feb 00:00 EST	1980780	319416	319674	511350	4
14 Feb 01:00 EST	1680820	266415	268518	429594	3
14 Feb 02:00 EST	1988123	320598	320862	513245	4
14 Feb 03:00 EST	1989027	320750	320997	513474	5
14 Feb 04:00 EST	1846467	296658	298544	477283	4
14 Feb 05:00 EST	1994929	321666	321969	515038	3
14 Feb 06:00 EST	1122055	177412	182165	291954	4
14 Feb 07:00 EST	1230917	195364	200032	319999	2
14 Feb 08:00 EST	1686943	271030	272703	436250	5
14 Feb 09:00 EST	1986730	320376	320634	512884	3
14 Feb 10:00 EST	1734486	278013	280594	448722	5
14 Feb 11:00 EST	1976516	318714	318996	510244	4
14 Feb 12:00 EST	1966204	317069	317299	507591	5
14 Feb 13:00 EST	794035	128042	128151	204987	1
Summary, Min, Max and Totals:					
	23,958,032	3,851,523	3,871,138	6,192,615	52

Metric	Description
Time	Time interval selected (yesterday, today, pick a day, pick a range).
MEGACO Packets	Total number of MGCP packets received during the interval.
MEGACO Adds	Total number of <i>Add</i> commands during the interval.
MEGACO Subtracts	Total number of <i>Subtract</i> commands during the interval.
MEGACO Modifys	Total number of <i>Modify</i> commands during the interval.
MEGACO Errors	Total number of MEGACO Errors registered during the interval.

H.323 Protocol Summary Report

The H.323 Protocol Summary Report, shown in the following figure, summarizes packet counts for H.323 including signaling, connection and channel control, setup and breakdown.

H.323 Protocol Summary Report for Thursday 08 September 2005 - Saturday 10 September 2005									
Time	H.323 Packets	H.225.0-RAS Packets	H.225.0-Q.93x Packets	H.245 Packets	H.225.0 Setup	H.225.0 CallProceeding	H.225.0 Alerting	H.225.0 Connect	H.225.0 ReleaseComplete
8 Sep 00:00 EDT	0	0	0	0	0	0	0	0	0
9 Sep 00:00 EDT	27	0	5	22	1	0	1	1	2
Summary Min, Max and Totals:									
	27	0	5	22	1	0	1	1	2

The following table lists and describes the metrics included in this report.

Metric	Description
Time	Time interval selected (yesterday, today, pick a day, pick a range).
H.323 Packets	The total number of H.323 packets received during the interval.
H.225.0-RAS Packets	The total number of Registration, Admission, Status message packets received during the interval.
H.225.0-Q.93x Packets	The total number of Setup/Breakdown message packets received during the selected interval.
H.245 Packets	The total number of media signaling packets received during the selected interval.

Metric	Description
H.225.0 Setup	The total number of packets used for call setup signaling between H.323 endpoints, received during the selected interval.
H.225.0 Call Proceeding	The total number of H.225.0 Call Proceeding message packets received during the selected interval.
H.225.0 Alerting	The total number of H.225.0 Alerting message packets received during the selected interval.
H.225.0 Connect	The total number of H.225.0 Connect message packets received during the selected interval.
H.225.0 Release Complete	The total number of H.225.0 Release Complete message packets received during the selected interval.

ISDN Protocol Summary Report

The ISDN Protocol Summary Report displays ISDN packet total along with key message packet counts that the total comprises.

ISDN Protocol Summary Report for Thursday 08 September 2005 - Saturday 10 September 2005										
Time	Packets	Receiver Readys	Setups	Call Proceeding	Alerts	Connects	Connect Acks	Disconnects	Releases	Release Completes
8 Sep 00:00 EDT	0	0	0	0	0	0	0	0	0	0
9 Sep 00:00 EDT	100	50	8	8	0	8	8	6	6	6
Summary Min, Max and Totals:										
	100	50	8	8	0	8	8	6	6	6

The following table lists and describes the metrics included in this report.

Metric	Description
Time	Time interval selected (yesterday, today, pick a day, pick a range).
Packets	The total number of ISDN packets received during the selected interval.
Receiver Readys	The total number of Receiver Ready messages received during the selected interval.
Setups	The total number of Setup messages received during the selected interval.
Call Proceeding	The total number of Call Proceeding messages received during the selected interval.
Alerts	The total number of Alert messages received during the selected interval.
Connects	The total number of Connect messages received during the selected interval.

Metric	Description
Connect Acks	The total number of Connect Acknowledgement messages received during the selected interval.
Disconnects	The total number of Disconnect messages received during the selected interval.
Releases	The total number of Release messages received during the selected interval.
Release Completes	The total number of Release Complete messages received during the selected interval.

SIGTRAN Protocol Summary Report

The SIGTRAN Protocol Summary Report displays SIGTRAN packet total along with key packet counts for SCTP, M3UA, M2PA, and M2PA ISUP.

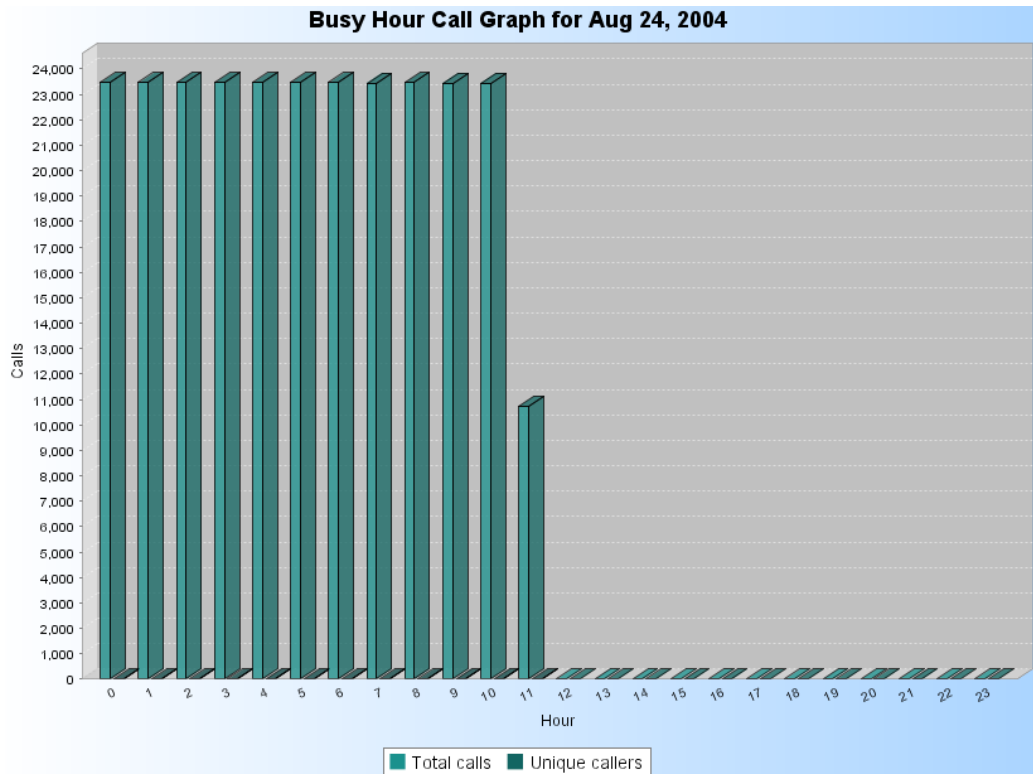
SIGTRAN Protocol Summary Report for Wednesday 15 March 2006														
Time	SCTP Packets	SCTP Abort	SCTP Error	M3UA Packets	M3UA Isup Packets	M3UA Mgmt Error	M3UA SSNM Packets	M3UA ASPSM Packets	M3UA ASPTM	M3UA RKM Packets	M2PA Packets	M2PA Isup Packets	M2PA Link Stat	M2PA User Data
15 Mar 00:00 EST	0	0	0	0	0	0	0	0	0	0	0	0	231	0
15 Mar 01:00 EST	0	0	0	0	0	0	0	0	0	60371	0	12	0	0
15 Mar 02:00 EST	29476	640	4637	0	0	0	0	0	0	0	0	0	0	0
15 Mar 03:00 EST	6621	0	0	0	0	0	0	0	0	0	0	0	0	0
15 Mar 04:00 EST	0	0	0	91924	0	0	0	0	119	0	0	0	0	0
15 Mar 05:00 EST	0	0	0	8979	0	0	0	0	0	0	0	10293	0	0
15 Mar 06:00 EST	0	0	1048	0	0	0	0	0	0	0	0	0	0	0
15 Mar 07:00 EST	0	0	0	0	0	6829	0	0	0	0	0	0	0	0
15 Mar 08:00 EST	0	0	6888	0	0	0	0	0	0	0	0	0	0	0
15 Mar 09:00 EST	0	0	0	0	0	0	0	0	0	0	0	0	69	0
15 Mar 10:00 EST	5362	0	0	0	0	23001	0	0	0	0	0	0	0	0
15 Mar 11:00 EST	0	0	0	0	0	0	0	0	0	0	0	0	0	0
15 Mar 12:00 EST	0	0	0	9584	0	6958	0	0	0	0	0	2310	0	0
15 Mar 13:00 EST	0	0	0	0	0	0	0	0	0	0	0	0	0	0
15 Mar 14:00 EST	0	0	0	0	0	0	0	0	0	0	0	0	0	0
15 Mar 15:00 EST	0	0	0	0	0	0	0	0	0	0	0	0	0	0
15 Mar 16:00 EST	0	0	0	0	0	0	0	0	0	0	0	0	0	0
15 Mar 17:00 EST	0	0	0	0	0	0	0	0	0	0	0	0	0	0
15 Mar 18:00 EST	0	0	0	0	0	0	0	0	0	0	0	0	0	0
15 Mar 19:00 EST	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Summary Min, Max and Totals:														
	45232	640	10988	110374	0	35943	0	0	119	0	0	12603	0	0

The following metrics are included in this report::

Metric	Description
Time	Time interval selected (yesterday, today, pick a day, pick a range).
SCTP Packets	The total number of SCTP packets received during the selected interval.
SCTP Abort	The total number of SCTP Aborts that occurred during the selected interval.
SCTP Errors	The total number of SCTP Errors packets received during the selected interval.
M3UA Packets	The total number of M3UA packets received during the selected interval.
M3UA ISUP Packets	The total number of M3UA ISUP packets received during the selected interval.
M3UA Mgnt Errors	The total number of M3UA Mgnt Errors received during the selected interval.
M3UA SSNM Packets	The total number of M3UA SSNM packets received during the selected interval.
M3UA ASPSM Packets	The total number of ISDN packets received during the selected interval.
M3UA ASPTM Packets	The total number of ISDN packets received during the selected interval.
M3UA RKM Packets	The total number of ISDN packets received during the selected interval.
M2PA Packets	The total number of ISDN packets received during the selected interval.
M2PA ISUP Packets	The total number of ISDN packets received during the selected interval.
M2PA Link Stat	The number of the MTP2 messages sent between M2PA peers to indicate link status.
M2PA User Data	The number of messages (data) sent from M2PA to a peer at another node (SCTP).

Busy-Hour Call Analysis Report

The BHCA Report, shown in the following figure, provides data on call volume, broken down hourly, for the current day.



The following table lists and describes the metrics included in this report.

Metric	Description
Total Calls	Total number of calls for this interval.
Unique Callers	Total number of unique callers for this interval.

Appendix A – Error Codes

The following table provides a listing of potential error codes for each of the VoIP protocols supported by Hammer XMS.

MGCP

MGCP Code	Meaning
400	Unspecified transient error.
401	The phone is already off hook.
402	The phone is already on hook.
403	Endpoint does not have sufficient resources at this time.
404	Insufficient bandwidth at this time.
405	The endpoint is restarting.
406	Transaction time-out
407	Transaction aborted.
409	Internal overload.
410	No endpoint available.
500	The endpoint is unknown.
501	The endpoint is not ready.
502	Endpoint doesn't have sufficient resources (permanent).
503	All of wildcard too complicated.
504	Unknown or unsupported command.
505	Unsupported RemoteConnectionDescriptor.
506	Unable to satisfy both LocalConnectionOptions and RemoteConnectionDescriptor.
507	Unsupported functionality.
508	Unknown or unsupported quarantine handling.
509	Error in RemoteConnectionDescriptor.
510	Unspecified protocol error was detected.
511	Command contained an unrecognized extension.

MGCP Code	Meaning
512	Gateway not equipped to detect a requested event.
513	Gateway is not equipped to generate one of the requested signals.
514	Gateway cannot send the specified announcement.
515	The transaction refers to an incorrect connection-id
516	Unknown call-id, or the incorrect call-id.
517	Unsupported or invalid mode.
518	Unsupported or unknown package.
519	Endpoint does not have a digit map.
520	The endpoint is restarting.
521	Endpoint redirected to another Call Agent.
522	No such event or signal.
523	Unknown action or illegal combination of actions.
524	Internal inconsistency in LocalConnectionOptions.
525	Unknown extension in LocalConnectionOptions.
526	Insufficient bandwidth.
527	Missing RemoteConnectionDescriptor.
528	Incompatible protocol version.
529	Internal hardware failure.
530	CAS signaling protocol error.
531	Failure of a grouping of trunks (e.g., facility failure).
532	Unsupported value(s) in LocalConnectionOptions.
533	Response too large.
534	Codec negotiation failure.
535	Packetization period not supported.
536	Unknown or unsupported Restart Method.
537	Unknown or unsupported digit map extension.
538	Event/signal parameter error
539	Invalid or unsupported command parameter.

MGCP Code	Meaning
540	Per endpoint connection limit exceeded.
541	Invalid or unsupported Local Connection Options.
900	Endpoint malfunctioning.
901	Endpoint taken out-of-service.
902	Loss of lower layer connectivity (e.g., downstream sync)
903	QoS resource reservation was lost.
904	Manual intervention.
905	Facility failure (e.g., DS-0 failure).
99996	Connecting Timeout
99997	Connected Timeout
99998	Disconnecting Timeout
99999	Closed by system

SIP

SIP Code	Meaning
400	Bad Request.
401	Unauthorized.
402	Payment Required.
403	Forbidden.
404	Not Found.
405	Method Not Allowed.
406	Not Acceptable.
407	Proxy Authentication Required.
408	Request Timeout.
409	Conflict.
410	Gone.
411	Length Required.
413	Request Entity Too Large.

SIP Code	Meaning
414	Request-URI Too Large.
415	Unsupported Media Type.
420	Bad Extension.
480	Temporarily not available.
481	Call Leg/Transaction Does Not Exist.
482	Loop Detected.
483	Too Many Hops.
484	Address Incomplete.
485	Ambiguous.
486	Busy Here.
500	Internal Server Error.
501	Not Implemented.
502	Bad Gateway.
503	Service Unavailable.
504	Gateway Time-out
505	Version Not Supported
600	Busy Everywhere.
603	Decline
604	Does Not Exist Anywhere
606	Not Acceptable
99996	Connecting Timeout
99997	Connected Timeout
99998	Disconnecting Timeout
99999	Closed by system

SS7

SS7 Code	Meaning
1	Unallocated number.
2	No route to network.
3	No route to destination.
4	Send special information tone.
5	Normal unspecified.
8	Preemption.
9	Preemption - Circuit Reserved.
16	Normal call clearing.
17	User busy.
18	No user responding.
19	No answer from the user.
20	Subscriber absent.
21	Call rejected.
22	Number changed.
23	ITU-Redirected ANSI-Unallocated destination number.
26	Non-selected user clearing.
27	Destination out of order.
28	Address incomplete.
29	Facility rejected.
31	Normal unspecified.
34	No circuit available.
38	Network out of order.
41	Temporary failure.
42	Switching equipment congestion.
43	Access information discarded.
44	Requested channel not available.
46	Precedence call blocked.

SS7 Code	Meaning
47	Resource unavailable.
50	Requested Facility Not Subscriber.
53	ITU-Outgoing calls barred within CUG.
55	ITU-incoming calls barred within CUG.
57	Bearer capability not authorized.
58	Bearer capability not presently.
62	Inconsistency.
63	Service/Option not available.
65	Bearer capability not implemented.
69	Requested facility not implemented.
70	Only restricted digital info capability available.
79	Service or option not implemented.
87	ITU-User not member of CUG.
88	Incompatible destination.
90	ITU-Non-existent CUG.
91	Invalid transit network selection.
95	Invalid message.
97	Message type non-existent.
99	Parameter non-existent.
102	Recovery of timer expiry.
103	ITU-Non-existent parameter passed on.
110	Message discarded - unrecognized parameter.
111	Protocol Error.
127	Interworking unspecified.
99996	Connecting Timeout
99997	Connected Timeout
99998	Disconnecting Timeout
99999	Closed by system

H.323

H.323 Code	Meaning
1	Unallocated (unassigned) number
2	No route to specified transit network
3	No route to destination
4	Send special information tone
5	Misdialed trunk prefix
6	Channel unacceptable
7	Call awarded - delivered in an established channel
8	Preemption
9	Preemption - circuit reserved for reuse
16	Normal call clearing
17	User busy
18	No user responding
19	No answer from user (user alerted)
20	Subscriber absent
21	Call rejected
22	Number changed
26	Non-selected user clearing
27	Destination out of order
28	Invalid number format (address incomplete)
29	Facility rejected
30	Response to STATUS INQUIRY
31	Normal - unspecified
34	No circuit/channel available
35	Call queued
38	Network out of order
39	Permanent frame mode connection out of service
40	Permanent frame mode connection operational

H.323 Code	Meaning
41	Temporary failure
42	Switching equipment congestion
43	Access information discarded
44	Requested circuit/channel not available
46	Precedence call blocked
47	Resource unavailable - unspecified
49	Quality of service not available
50	Requested facility not subscribed
52	Outgoing calls barred
53	Outgoing calls barred within CUG
54	Incoming calls barred
55	Incoming calls barred within CUG
57	Bearer capability not authorized
58	Bearer capability not presently available
62	Inconsistency in designated outgoing access information and subscriber class
63	Service or option not available - unspecified
65	Bearer capability not implemented
66	Channel type not implemented
69	Requested facility not implemented
70	Only restricted digital information bearer capability is available
79	Service or option not implemented - unspecified
81	Invalid call reference value
82	Identified channel does not exist
83	A suspended call exists but this call identity does not
84	Call identity in use
85	No call suspended

H.323 Code	Meaning
86	Call having the requested call identity has been cleared
87	User not a member of CUG
88	Incompatible destination
90	Non-existent CUG
91	Invalid transit network selection
95	Invalid message - unspecified
96	Mandatory information element is missing
97	Message type non-existent or not implemented
98	Message not compatible with call state or message type non-existent or not implemented
99	Information element/parameter non-existent or not implemented
100	Invalid information element contents
101	Message not compatible with call state
102	Recovery on timer expired
103	Parameter non-existent or not implemented - passed on
110	Message with unrecognized parameter - discarded
111	Protocol error - unspecified
127	Interworking - unspecified
500	Called Party Not Registered 16600
501	Invalid Permission
502	Request Denied
503	Undefined Reason
504	Caller Not Registered
505	Route Call To Gatekeeper
506	Invalid EndpointIdentifier
507	Resource Unavailable
508	Security Denial

H.323 Code	Meaning
509	QOS Control Not Supported
510	Incomplete Address 16610
511	Aliases Inconsistent
512	Route Call ToSCN
513	Exceeds Call Capacity
514	Collect Destination
515	Collect PIN
516	Generic DataReason
517	Needed Feature Not Supported
518	Security Wrong SyncTime
519	Security Replay
520	Security Wrong GeneralID 16620
521	Security Wrong SendersID
522	Security Integrity Failed
523	Security WrongOID
524	Security DH mismatch
525	NoRoute To Destination
526	Unallocated Number
527	No Bandwidth
528	Gatekeeper Resources
529	Unreachable Destination
530	Destination Rejection 16630
531	Invalid Revision
532	No Permission
533	Unreachable Gatekeeper
534	Gateway Resources
535	Bad Format Address
536	Adaptive Busy

H.323 Code	Meaning
537	InConf
538	Undefined Reason
539	Facility Call Deflection
540	Security Denied 16640
541	Security Wrong Sync Time
542	Security Replay
543	Security Wrong General ID
544	Security Wrong Senders ID
545	Security Integrity Failed
546	Security Wrong OID
547	Security DH mismatch
548	Security CertificateE xpired
549	Security Certificate Date Invalid
550	Security Certificate Revoked 16650
551	Security Certificate Not Readable
552	Security Certificate Signature Invalid
553	Security Certificate Missing
554	Security Certificate Incomplete
555	Security Unsupported Certificate Alg OID
556	Security Unknown CA
557	Called Party Not Registered
558	Caller Not Registered
559	New Connection Needed
560	Non Standard Reason 16660
561	Replace With Conference Invite
562	Generic Data Reason
563	Needed Feature Not Supported
564	Tunnelled Signalling Rejected

H.323 Code	Meaning
565	Invalid CID
566	Hop Count Exceeded
567	Forced Drop
568	Normal Drop
569	Undefined Reason
99996	Connecting Timeout
99997	Connected Timeout
99998	Disconnecting Timeout
99999	Closed by system

ISDN

ISDN Code	Meaning
1	Unallocated (unassigned) number
2	No route to specified transit network
3	No route to destination
4	Send special information tone
5	Misdialed trunk prefix
6	Channel unacceptable
8	Preemption
9	Preemption - circuit reserved for reuse
16	Normal call clearing
17	User busy
18	No user responding
19	No answer from user (user alerted)
20	Subscriber absent
21	Call rejected
22	Number changed

ISDN Code	Meaning
23	ITU-Redirected ANSI-Unallocated destination number
26	Non-selected user clearing
27	Destination out of order
28	Invalid number format (address incomplete)
29	Facility rejected
30	Response to STATUS INQUIRY
31	Normal - unspecified
34	No circuit/channel available
38	Network out of order
41	Temporary failure
42	Switching equipment congestion
43	Access information discarded
44	Requested circuit/channel not available
45	Call preempted
46	Precedence call blocked
47	Resource unavailable - unspecified
50	Requested facility not subscribed
52	Outgoing calls barred
53	Outgoing calls barred within CUG
54	Incoming calls barred
55	Incoming calls barred within CUG
57	Bearer capability not authorized
58	Bearer capability not presently available
62	Inconsistency in designated outgoing access information and subscriber class
63	Service or option not available - unspecified
65	Bearer capability not implemented
66	Channel type not implemented

ISDN Code	Meaning
69	Requested facility not implemented
70	Only restricted digital information bearer capability is available
79	Service or option not implemented - unspecified
81	Invalid call reference value
82	Identified channel does not exist
87	User not a member of CUG
88	Incompatible destination
90	Non-existent CUG
91	Invalid transit network selection
95	Invalid message - unspecified
96	Mandatory information element is missing
97	Message type non-existent or not implemented
98	Message not compatible with call state or message type non-existent or not implemented
99	Information element/parameter non-existent or not implemented
100	Invalid information element contents
101	Message not compatible with call state
102	Recovery on timer expired
103	Parameter non-existent or not implemented - passed on
110	Message with unrecognized parameter - discarded
111	Protocol error - unspecified
127	Interworking - unspecified
99996	Connecting Timeout
99997	Connected Timeout
99998	Disconnecting Timeout
99999	Closed by System

MEGACO / H.248

H.248 Code	Meaning
400	Syntax error in message
401	Protocol Error
402	Unauthorized
403	Syntax Error in TransactionRequest
406	Version Not Supported
410	Incorrect identifier
411	The transaction refers to an unknown ContextID
412	No ContextIDs available
413	Number of transactions in message exceeds maximum
421	Unknown action or illegal combination of actions
422	Syntax Error in Action
430	Unknown TerminationID
431	No TerminationID matched a wildcard
432	Out of TerminationIDs or No TerminationID available
433	TerminationID is already in a Context
434	Max number of Terminations in a Context exceeded
435	Termination ID is not in specified Context
440	Unsupported or unknown Package
441	Missing Remote or Local Descriptor
442	Syntax Error in Command
443	Unsupported or Unknown Command
444	Unsupported or Unknown Descriptor
445	Unsupported or Unknown Property
446	Unsupported or Unknown Parameter
447	Descriptor not legal in this command
448	Descriptor appears twice in a command

H.248 Code	Meaning
449	Unsupported or Unknown Parameter or Property Value
450	No such property in this package
451	No such event in this package
452	No such signal in this package
453	No such statistic in this package
454	No such parameter value in this package
455	Property illegal in this Descriptor
456	Property appears twice in this Descriptor
457	Missing parameter in signal or event
458	Unexpected Event / Request ID
459	Unsupported or Unknown Profile
460	Unable to set statistic on stream
471	Implied Add for Multiplex failure
500	Internal software failure in the MG
501	Not Implemented
502	Not ready
503	Service Unavailable
504	Command Received from unauthorized entity
505	Transaction Request received before a ServiceChange reply has been received
506	Number of TransactionPendings Exceeded
510	Insufficient resources
512	Media Gateway unequipped to detect requested Event
513	Media Gateway unequipped to generate requested Signals
514	Media Gateway cannot send the specified announcement
515	Unsupported Media Type

H.248 Code	Meaning
517	Unsupported or invalid mode
518	Event buffer full
519	Out of space to store digit map
520	Digit Map undefined in the MG
521	Termination is 'Service Changing'
522	Functionality Requested in Topology Triple Not Supported
526	Insufficient bandwidth
529	Internal hardware failure in MG
530	Temporary Network failure
531	Permanent Network failure
532	Audited Property or Statistic or Event or Signal does not exist
533	Response exceeds maximum transport PDU size
534	Illegal write of read only property
540	Unexpected initial hook state
542	Command is not allowed on this termination
581	Does Not Exist
600	Illegal syntax within an announcement specification
601	Variable type not supported
602	Variable value out of range
603	Category not supported
604	Selector type not supported
605	Selector value not supported
606	Unknown segment ID
607	Mismatch between play specification and provisioned data
608	Provisioning error
609	Invalid offset
610	No free segment IDs

H.248 Code	Meaning
611	Temporary segment not found
612	Segment in use
613	ISP port limit overrun
614	No modems available
615	Calling number unacceptable
616	Called number unacceptable
900	Service Restored
901	Cold Boot
902	Warm Boot
903	MGC Directed Change
904	Termination malfunctioning
905	Termination taken out of service
906	Loss of lower layer connectivity (e.g. downstream sync)
907	Transmission Failure
908	MG Impending Failure
909	MGC Impending Failure
910	Media Capability Failure
911	Modem Capability Failure
912	Mux Capability Failure
913	Signal Capability Failure
914	Event Capability Failure
915	State Loss
916	Packages Change
917	Capabilities Change
918	Cancel Graceful
919	Warm Failover
920	Cold Failover
99996	Connecting Timeout

H.248 Code	Meaning
99997	Connected Timeout
99998	Disconnecting Timeout
99999	Closed by System

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